Car Crash Business Report

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Exploratory Data Analysis

Problem Definition

Car crashes remain one of the leading causes of injury and death worldwide, and understanding the factors that influence survival is essential for improving road safety. Despite advancements in vehicle technology, the likelihood of surviving a crash depends on a complex interaction of factors such as **impact speed, use of safety features (seatbelts, airbags), type of collision, occupant demographics, and vehicle age**.

The Department of Road Transport has reported a **15% year-on-year increase in crashes in urban areas**, underlining the urgency of addressing this issue. While accident causes are recorded after crashes occur, there is currently limited ability to **predict survival outcomes in advance** or to identify which factors most strongly drive fatalities.

The goal of this project is to use **five years of historical crash data** to:

1. **Analyze patterns and relationships** between crash factors and survival outcomes.
2. **Build predictive models** that estimate the probability of survival (or death) in a crash.
3. **Identify the most critical factors** influencing survival, enabling the Department to design better safety regulations and encourage manufacturers and road users to adopt life-saving measures.

Ultimately, this analysis aims to bridge the gap between raw crash records and actionable insights that can **reduce fatalities, improve vehicle design standards, and guide enforcement policies**.

Data Description

The data contains the different attributes of car crashes, with the outcome variable being whether the occupant was deceased during the crash or not. The detailed data dictionary is given below.

Data Dictionary:

- **caseid:** character, created by pasting together the population sampling unit, the case number, and the vehicle number. Within each year, use this to uniquely identify the vehicle.
- **speed_range:** factor with levels (estimated impact speeds) 1-9 km/h, 10-24 km/h, 25-39 km/h, 40-54 km/h, 55+ km/h
- **weight:** Observation weights, albeit of uncertain accuracy, are designed to account for varying sampling probabilities. (The inverse probability weighting estimator can be used to demonstrate causality when the researcher cannot conduct a controlled experiment but has observed data to model)
- **seatbelt:** a factor with levels none or belted
- **frontal_impact:** a numeric vector; 0 = non-frontal, 1=frontal impact
- **sex:** a factor with levels f: Female or m: Male
- **age_of_occ:** age of occupant in years
- **year_of_acc:** year of accident
- **model_year:** Year of model of vehicle; a numeric vector
- **airbag:** Did one or more (driver or passenger) airbag(s) deploy? This factor has levels deploy, nodeploy, and unavail
- **occ_role:** a factor with levels driver or pass: passenger

- **deceased:** the target variable with levels no (survived) or yes (not survived / deceased)

Data Description and Initial Observations

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
caseid	11217	6099	0.604872685	8	NaN	NaN	NaN	NaN	NaN	NaN	NaN
speed_range	11217	5	10-24 km/h	5414	NaN	NaN	NaN	NaN	NaN	NaN	NaN
weight	11217.00000	NaN	NaN	NaN	431.40531	1406.20294	0.00000	28.29200	82.19500	324.05600	31694.04000
seatbelt	11217	2	belted	7849	NaN	NaN	NaN	NaN	NaN	NaN	NaN
frontal_impact	11217.00000	NaN	NaN	NaN	0.64402	0.47883	0.00000	0.00000	1.00000	1.00000	1.00000
sex	11217	2	m	6048	NaN	NaN	NaN	NaN	NaN	NaN	NaN
age_of_occ	11217.00000	NaN	NaN	NaN	37.42765	18.19243	16.00000	22.00000	33.00000	48.00000	97.00000
year_of_acc	11217.00000	NaN	NaN	NaN	2001.10324	1.05681	1997.00000	2001.00000	2001.00000	2002.00000	2002.00000
model_year	11217.00000	NaN	NaN	NaN	1994.17794	5.65870	1953.00000	1991.00000	1995.00000	1999.00000	2003.00000
airbag	11217	3	deploy	4365	NaN	NaN	NaN	NaN	NaN	NaN	NaN
occ_role	11217	2	driver	8786	NaN	NaN	NaN	NaN	NaN	NaN	NaN
deceased	11217	2	no	10037	NaN	NaN	NaN	NaN	NaN	NaN	NaN
veh_usage_duration	11217.00000	NaN	NaN	NaN	6.92529	5.49315	-1.00000	2.00000	6.00000	10.00000	49.00000

Figure 1: Statistical Summary

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11217 entries, 0 to 11216
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   caseid          11217 non-null  object
 1   speed_range     11217 non-null  object
 2   weight          11217 non-null  float64
 3   seatbelt        11217 non-null  object
 4   frontal_impact  11217 non-null  int64
 5   sex             11217 non-null  object
 6   age_of_occ      11217 non-null  int64
 7   year_of_acc     11217 non-null  int64
 8   model_year      11217 non-null  int64
 9   airbag          11217 non-null  object
 10  occ_role        11217 non-null  object
 11  deceased        11217 non-null  object
dtypes: float64(1), int64(4), object(7)
memory usage: 1.0+ MB
```

Figure 2: Data Overview

The dataset contains **11,217 records** with **12 attributes**, covering five years of historical crash data. Each record corresponds to a unique vehicle and occupant combination in a crash, identified by the variable `caseid`.

Dataset dimensions: (11,217 rows × 12 columns)

Variables include:

- **Crash dynamics**: `speed_range` (categorical, ordered bins), `frontal_impact` (binary: 0 = non-frontal, 1 = frontal).
- **Safety features**: `seatbelt` (none/belted), `airbag` (deploy/nodeploy/unavail).
- **Occupant details**: `sex` (male/female), `age_of_occ` (integer, 16–97 years), `occ_role` (driver/passenger).
- **Vehicle characteristics**: `model_year` (1953–2003), `year_of_acc` (1997–2002), from which a derived variable `vehicle_age` can be computed.
- **Sampling weight**: `weight` (0–31,694), used to adjust for sampling probabilities.
- **Outcome variable**: `deceased` (yes/no), the target for prediction.

Data type summary:

- 7 categorical/object variables: `caseid`, `speed_range`, `seatbelt`, `sex`, `airbag`, `occ_role`, `deceased`.
- 4 numeric variables: `weight` (float), `frontal_impact`, `age_of_occ`, `year_of_acc`, `model_year` (integers).

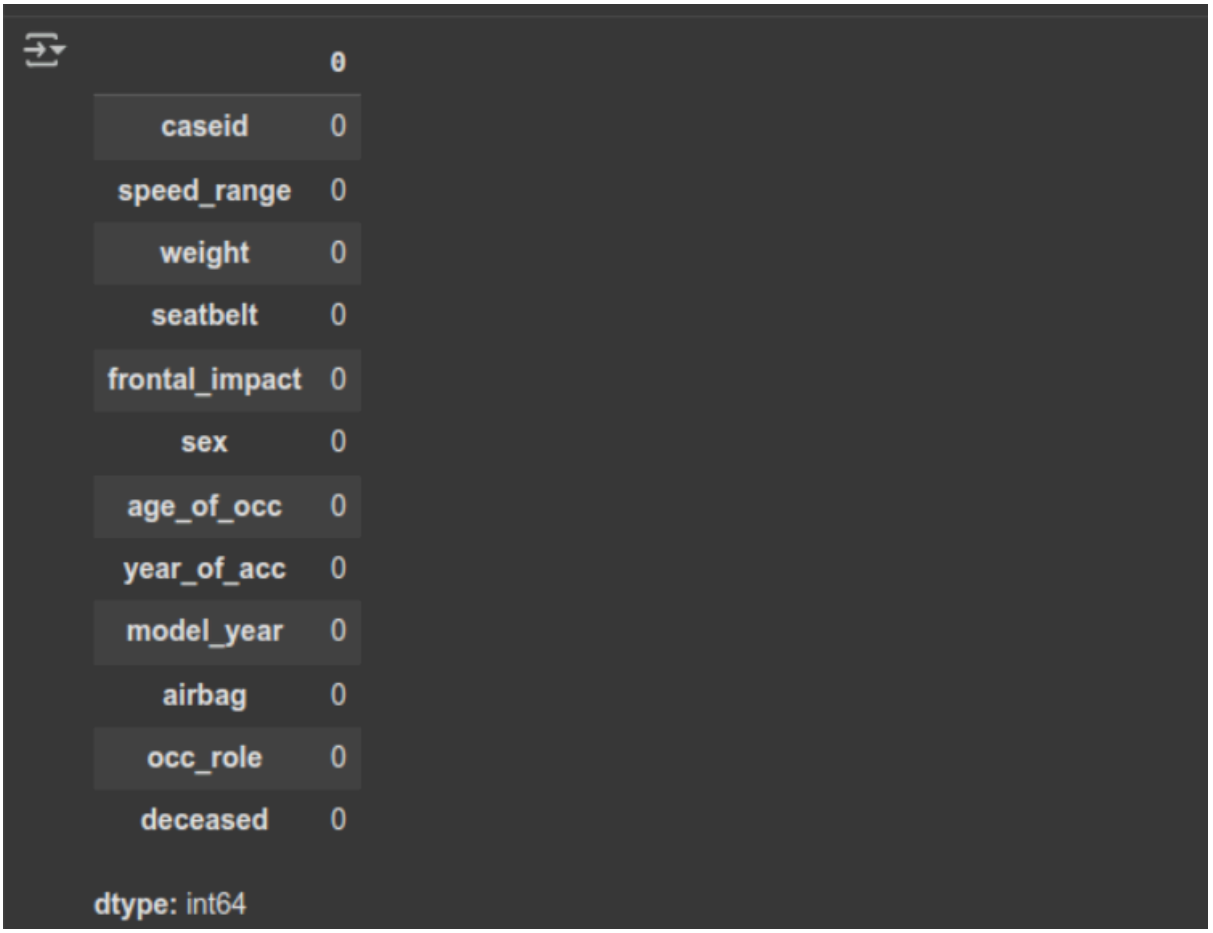
Statistical summary (selected observations):

- **Occupant age** ranges from 16 to 97, with a mean of ~37 years.
- **Accident years** span 1997–2002, while **vehicle model years** span 1953–2003. This implies some very old vehicles are included, as well as relatively modern ones.
- **Weights** vary widely (0–31,694), with a highly skewed distribution — the median is only ~82, suggesting a few cases carry very large weights.
- **Frontal impacts** account for about 64% of all crashes.

Initial checks:

- No missing values are reported for any of the 12 columns.
- Logical consistency needs to be checked (e.g., `model_year` should not exceed `year_of_acc`; any violations would need correction).
- Some outliers exist in `vehicle_age` because of very old model years (1950s and 1960s vehicles still in crashes). These will be handled during preprocessing.

Missing Value Check



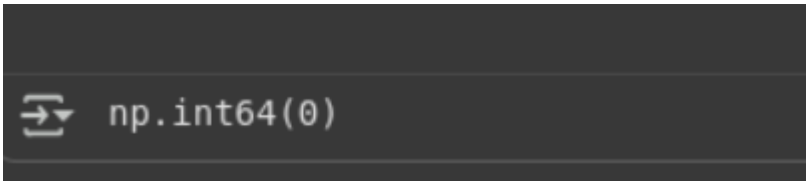
	0
caseid	0
speed_range	0
weight	0
seatbelt	0
frontal_impact	0
sex	0
age_of_occ	0
year_of_acc	0
model_year	0
airbag	0
occ_role	0
deceased	0

dtype: int64

Figure 3: Missing value check

A full null-value audit confirmed that the dataset has **no missing values** across any of the 12 variables. This ensures completeness of the data and reduces the need for imputation during preprocessing.

Duplicate Values



```
np.int64(0)
```

Figure 4: Duplicate value check

A duplicate check on the dataset returned **zero duplicate records**. Each crash case (`caseid`) is unique, which means the dataset has integrity at the row level and is reliable for analysis.

Creation of `veh_usage_duration` Variable

To capture the effect of vehicle aging, we engineered a new variable:

- **`veh_usage_duration = year_of_acc - model_year`**

This variable represents the number of years a vehicle had been in use at the time of the crash. The rationale is that **older vehicles are more likely to lack modern safety features and structural reinforcements**, which may increase fatality risk. Including this feature allows the models to quantify how vehicle age contributes to survival outcomes and supports actionable insights such as policies for **fleet renewal or mandatory safety upgrades**.

Univariate Analysis

1. deceased

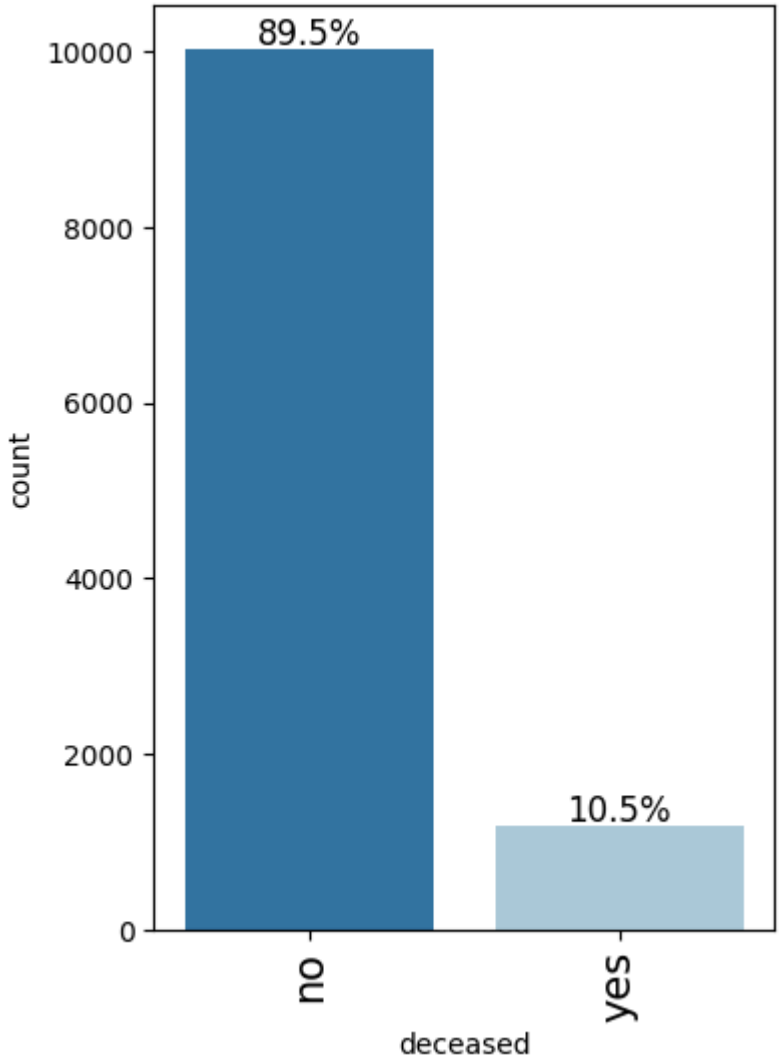


Fig.5: Barplot of deceased

The target variable **`deceased`** indicates whether the occupant survived the crash. The distribution is highly imbalanced:

- **89.5%** of cases correspond to survivors (`no`).
- **10.5%** correspond to fatalities (`yes`).

This imbalance shows that the majority of occupants survive crashes, while a smaller but significant proportion results in death. From a modeling perspective, this imbalance is important because a naive model could achieve nearly 90% accuracy by always predicting survival, but would fail to capture the high-risk cases that matter most.

Implication: The evaluation metric must focus on the **recall of the "deceased" class** (sensitivity) to ensure that the model identifies as many high-risk cases as possible.

2. weight

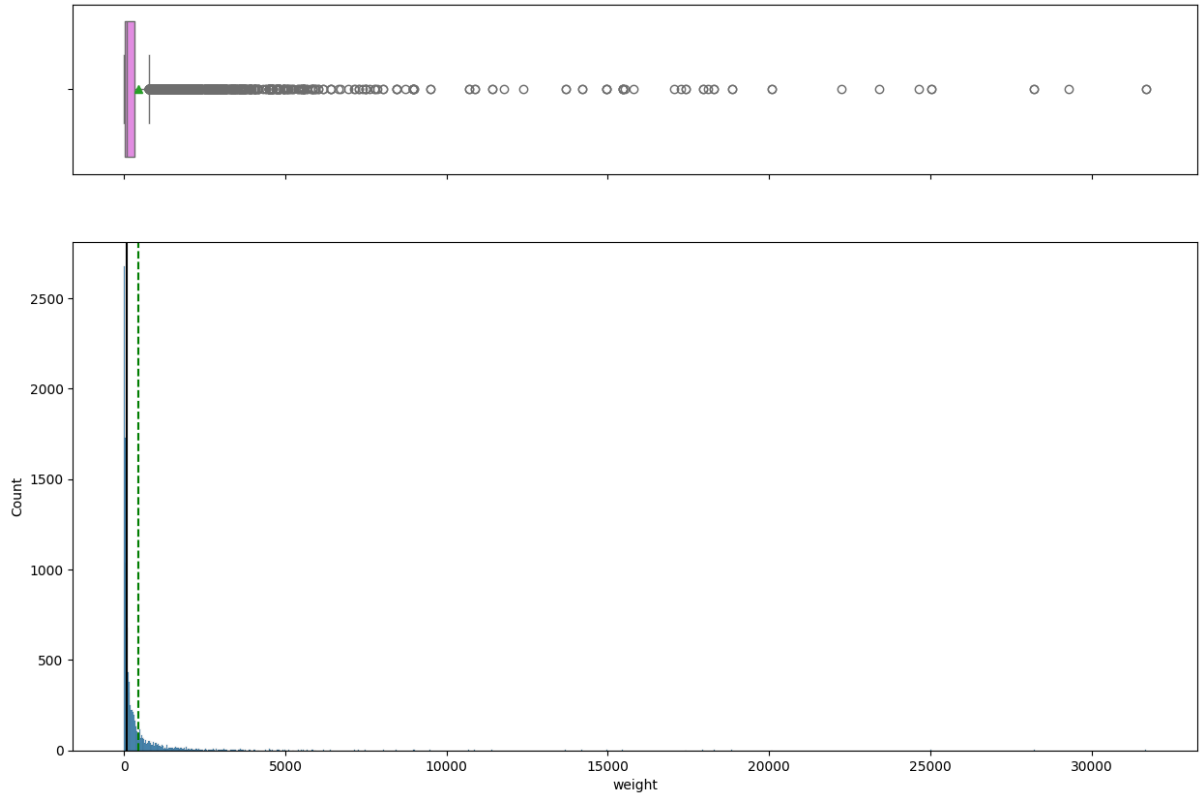


Fig.6: Histogram and Boxplot of weight

The `weight` variable, designed to adjust for varying sampling probabilities, shows an **extremely right-skewed distribution**:

- The **median weight** is only ~82, while the maximum extends to over **31,000**.
- The histogram shows the vast majority of cases clustered at the low end, with a long tail of high-weight outliers.
- The boxplot further confirms the presence of numerous extreme outliers, though these are expected because weights are a product of the sampling design, not measurement error.

Implication:

The skew indicates that a small number of records represent a disproportionately large share of the population. Rather than treating these values as errors, they will be retained and used carefully during **weighted summaries and modeling**, while unweighted analyses will be explicitly labeled to avoid misinterpretation.

3. age_of_occ

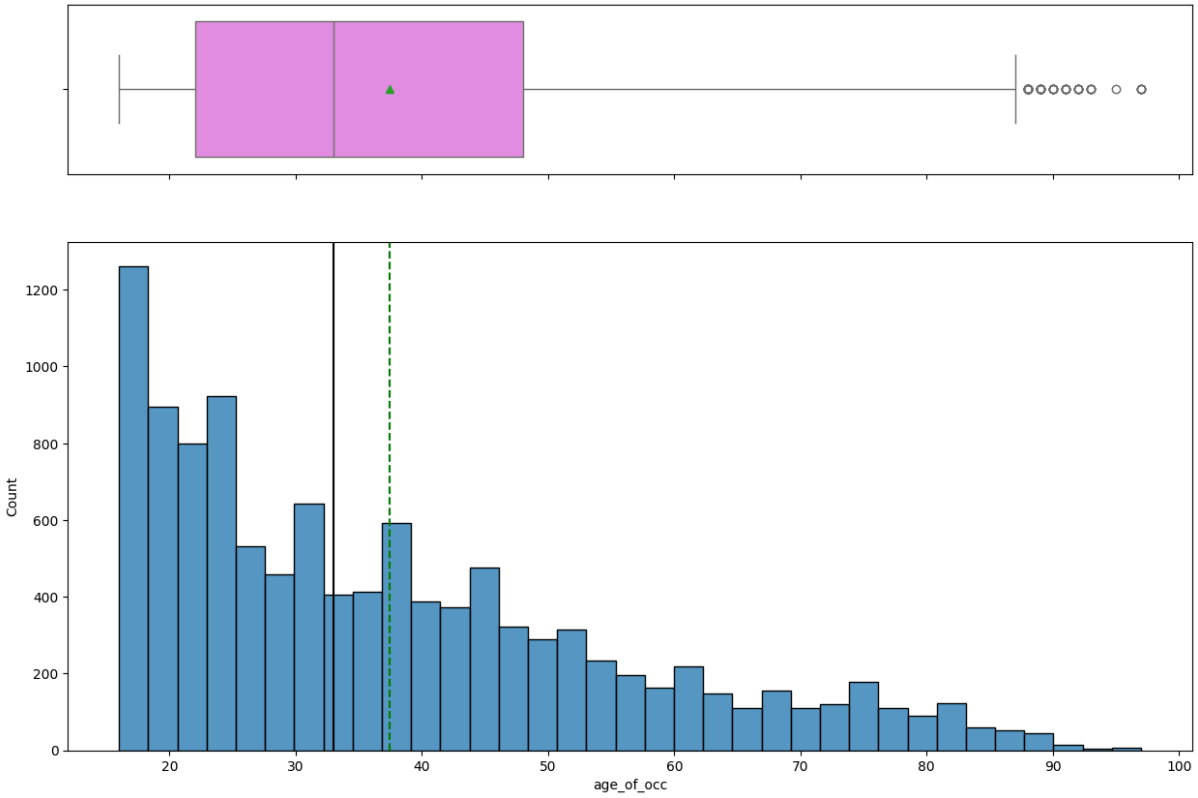


Fig.7: Histogram and Boxplot of age_of_occ

The ages of occupants involved in crashes range from **16 to 97 years**, with a mean of ~37 years and a median of 33 years.

- The histogram shows a **right-skewed distribution** with a large concentration of younger occupants in their late teens and 20s.
- The frequency steadily decreases with age, though crashes still occur across all age groups, including elderly occupants.
- The boxplot reveals a relatively wide interquartile range (22–48 years), and a few outliers at very advanced ages (above ~85 years).

Implication:

Younger age groups represent the majority of crash cases, but older individuals are still significantly present. From a safety perspective, it will be important to test whether **extremes of age** (very young or very old) correspond to higher fatality risk once combined with other factors such as speed and restraint use.

4. speedTable of Tables_range

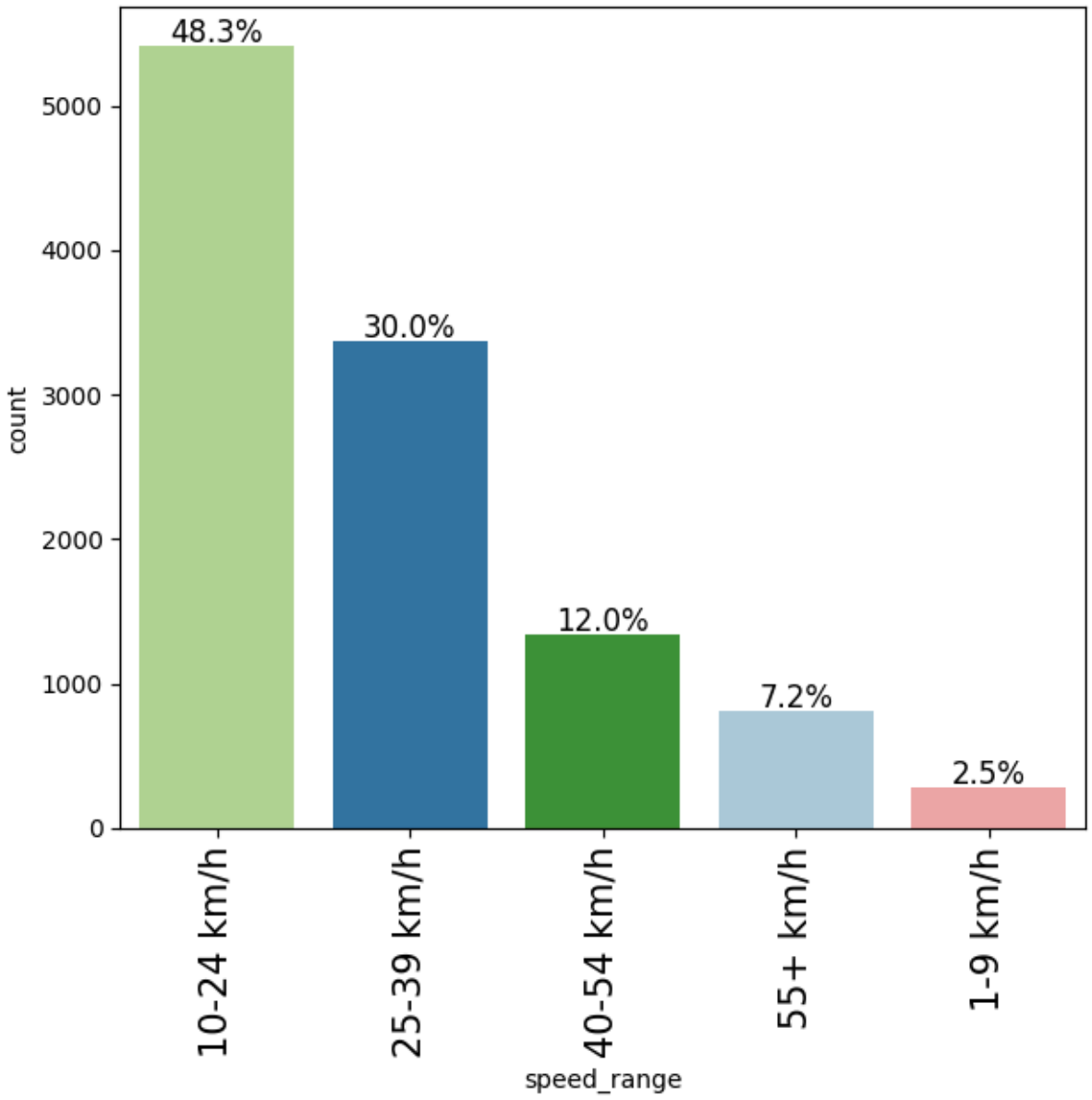


Fig.8: Bargraph for speed_range

Impact speed is grouped into ordered categories: *1–9 km/h, 10–24 km/h, 25–39 km/h, 40–54 km/h, 55+ km/h*.

- The largest share of crashes occur in the **10–24 km/h range (48.3%)**, followed by **25–39 km/h (30.0%)**.
- Higher-speed crashes are much less frequent: **40–54 km/h (12.0%)** and **55+ km/h (7.2%)** together account for under one-fifth of all cases.
- Very low-speed crashes (*1–9 km/h*) make up only **2.5%** of cases.

Implication:

Most crashes in the dataset occur at **low to moderate speeds**, which may reflect urban traffic conditions. However, even though higher-speed crashes are less frequent, they are expected to be associated with significantly higher fatality risk, an important relationship to confirm in the bivariate and modeling stages.

5. airbag

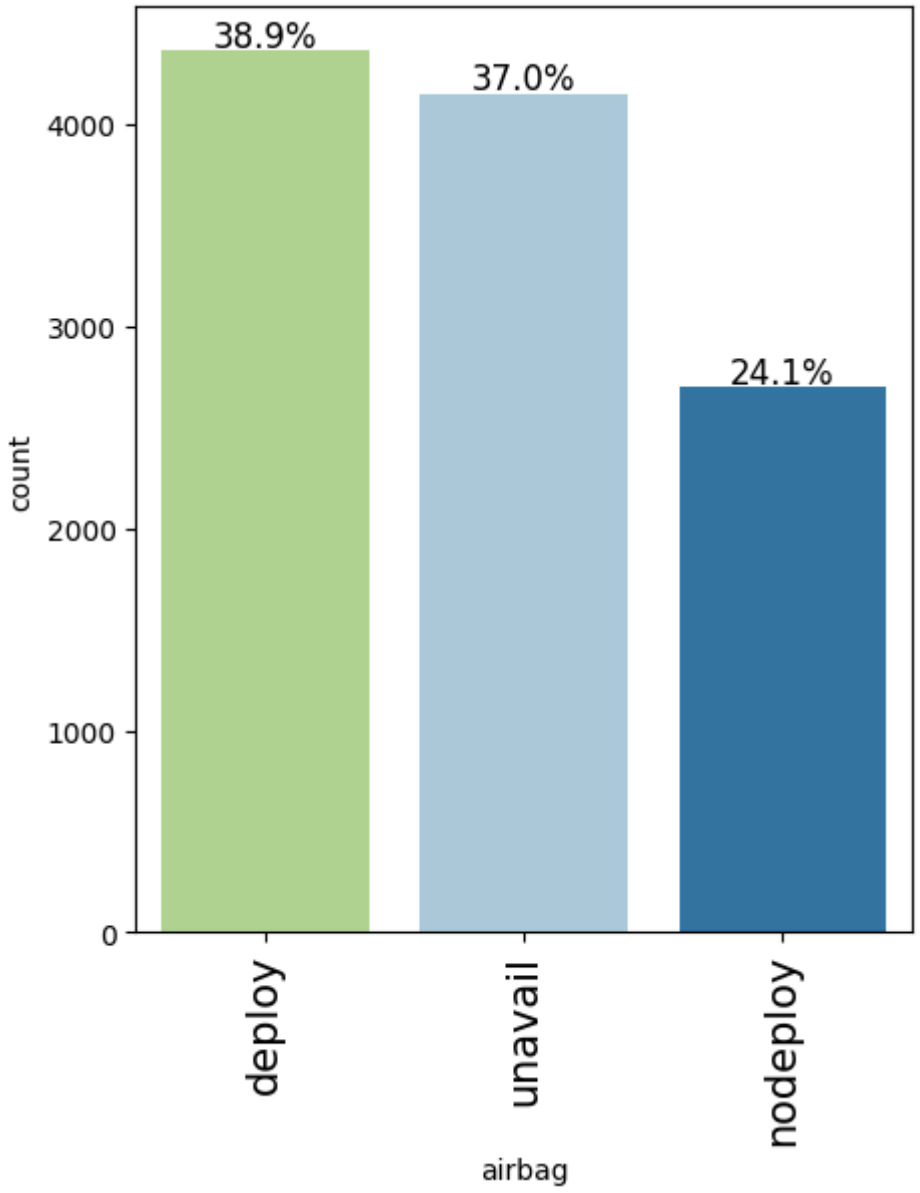


Fig.9: Bargraph of airbag

The dataset records whether airbags were deployed during the crash, not deployed, or unavailable:

- **Deploy:** 38.9% of cases
- **Unavailable:** 37.0% of cases
- **No deploy:** 24.1% of cases

The fact that nearly **four in ten records are marked as “unavail”** indicates either older vehicles without airbags or missing system information. This category is not a data gap but an informative level that reflects differences in vehicle technology.

Implication:

The deployment of airbags is expected to play a critical role in reducing fatalities, particularly in frontal impacts. The large share of unavailable cases underscores the importance of vehicle model year and usage duration in explaining survival outcomes.

6. seatbelt

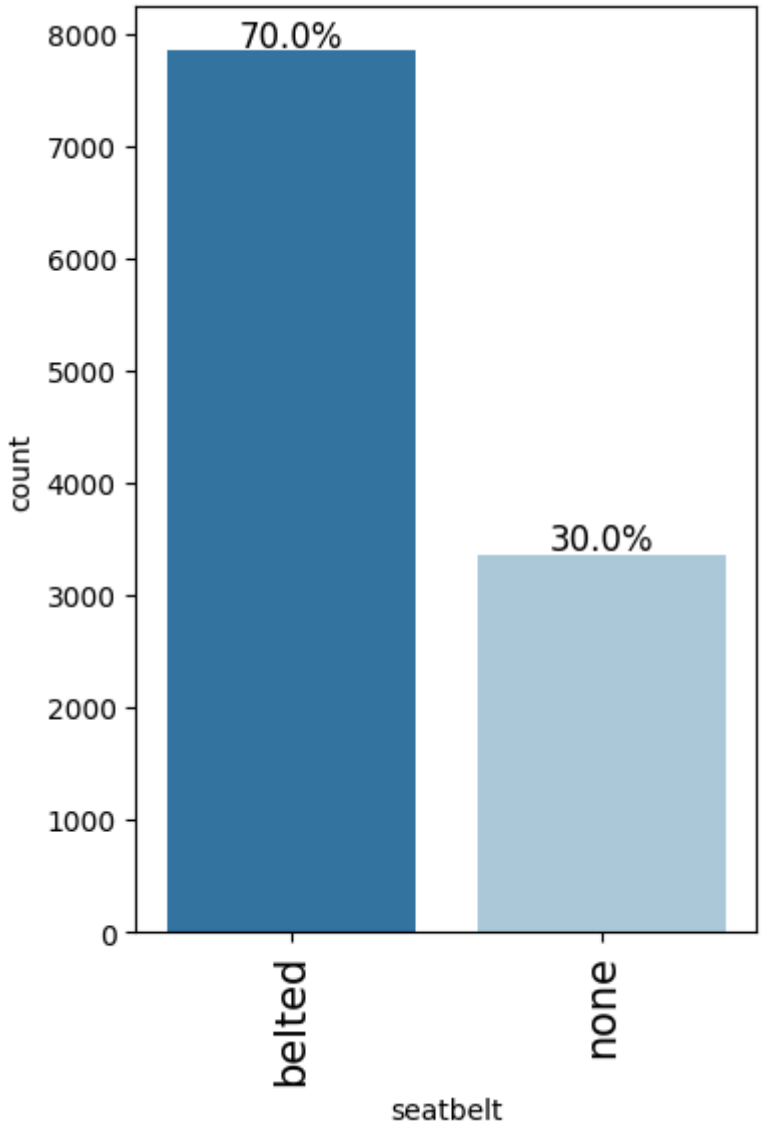


Fig.10: Bargraph for seatbelt

Seatbelt use among crash occupants is split between two categories:

- **Belted:** 70.0%
- **None:** 30.0%

This means nearly one in three occupants were not wearing a seatbelt at the time of the crash, a significant share considering the proven protective effect of seatbelts.

Implication:

Seatbelt usage is high but not universal, and the 30% without seatbelts represent a vulnerable group. Given that restraint systems are among the most effective safety interventions, we expect seatbelt non-use to be strongly associated with higher fatality rates, especially at higher speeds and during frontal impacts.

7. frontal_impact

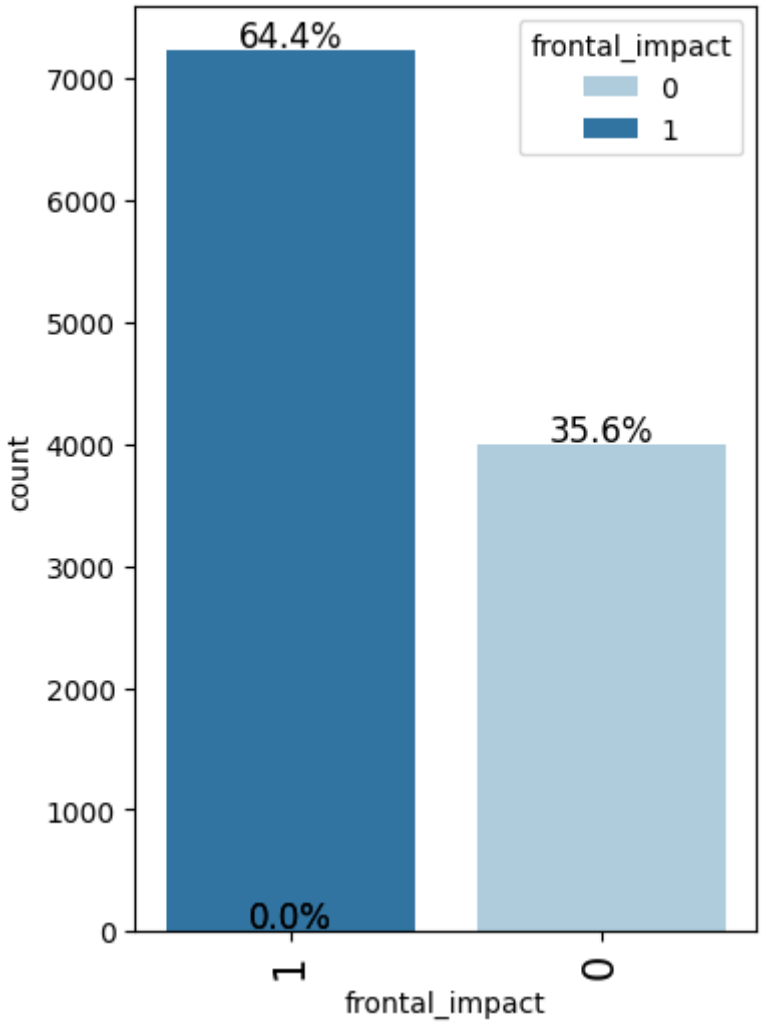


Fig.11: Bargraph for frontal_impact

Crashes are categorized as either **frontal (1)** or **non-frontal (0):**

- **Frontal impacts:** 64.4% of cases
- **Non-frontal impacts:** 35.6% of cases

This confirms that the majority of crashes in the dataset are frontal collisions. Since frontal impacts typically involve greater force transfer to occupants, they are a critical factor in survival analysis.

Implication:

The predominance of frontal impacts makes it essential to examine how safety features such as **seatbelts** and **airbag deployment** interact with this crash type. We expect that restraint systems will show their strongest protective effects in this category.

8. sex

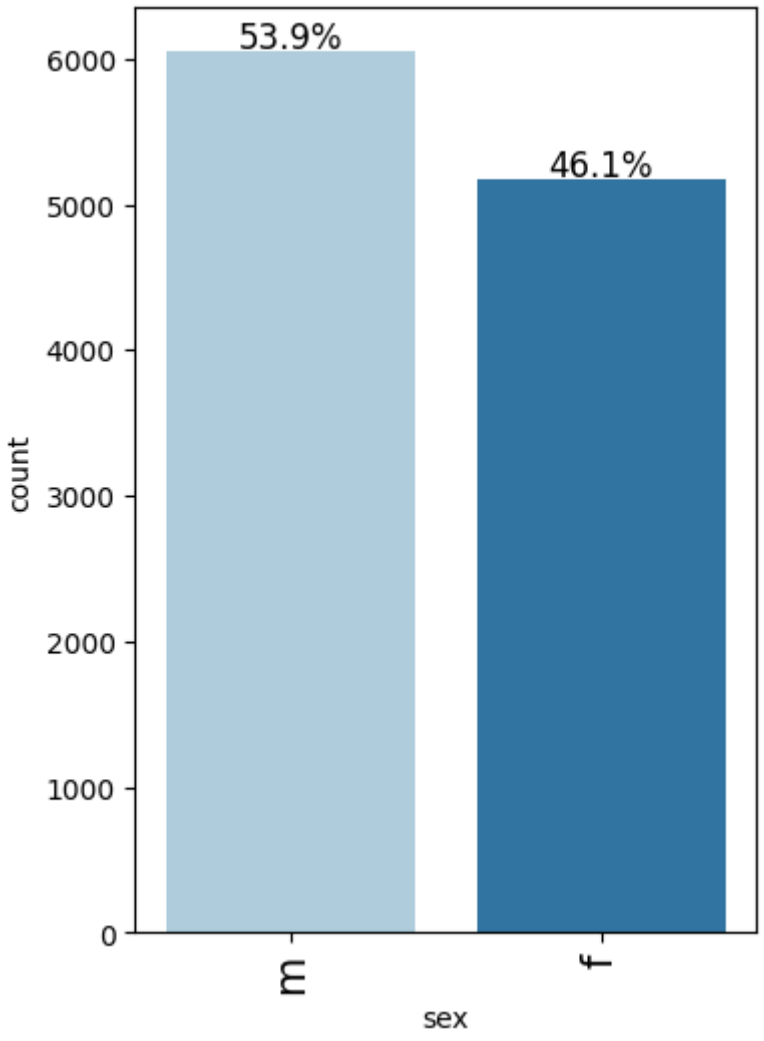


Fig.12: Bargraph for sex

The dataset includes both male and female occupants:

- **Male (m): 53.9%**
- **Female (f): 46.1%**

This indicates a slightly higher representation of males in crash records, which may reflect broader driving exposure patterns.

Implication:
Sex itself is unlikely to be a direct determinant of survival, but it may interact with other factors such as **occupant role** (driver vs passenger) or **safety feature usage**. Including it in the analysis helps confirm whether differences in fatality risk arise from demographic patterns or behavioral factors.

9. model_year

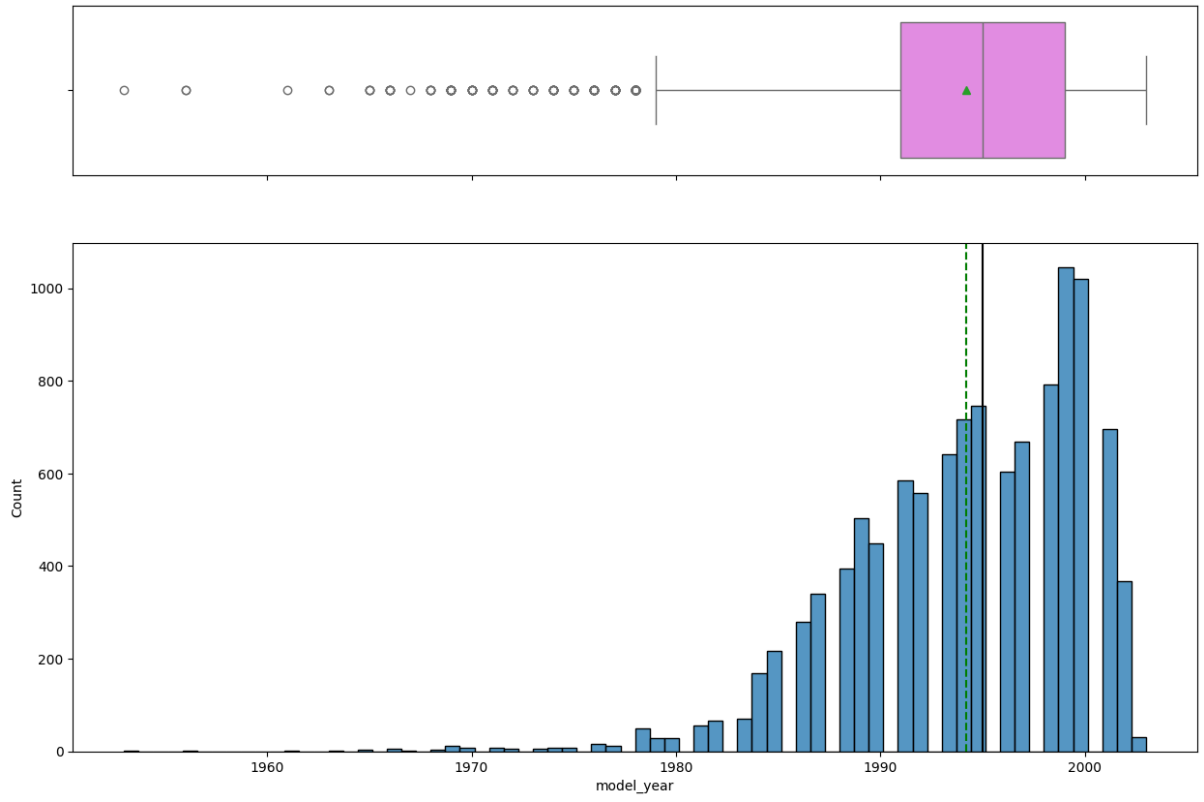


Fig.13: Histogram & boxplot for model_year

The vehicles in the dataset span a very wide range of model years, from **1953 to 2003**, with most concentrated in the **late 1980s to early 2000s**.

- The histogram shows very few vehicles from before 1970, followed by a steady increase in frequency through the 1980s and a peak around the **mid-to-late 1990s**.
- The median model year is **1995**, and the interquartile range falls between **1991 and 1999**, reflecting a dataset dominated by relatively modern vehicles for the study period.
- The boxplot reveals outliers in the form of very old vehicles (1950s–1960s), which represent a small but notable minority.

Implication:

Vehicle model year acts as a proxy for **technology level and safety features** (e.g., airbag availability). Older vehicles may lack modern protective systems, increasing fatality risk. By combining `model_year` with `year_of_acc`, the engineered variable `veh_usage_duration` will provide further insight into the effect of vehicle age at the time of the crash.

10. occ_role

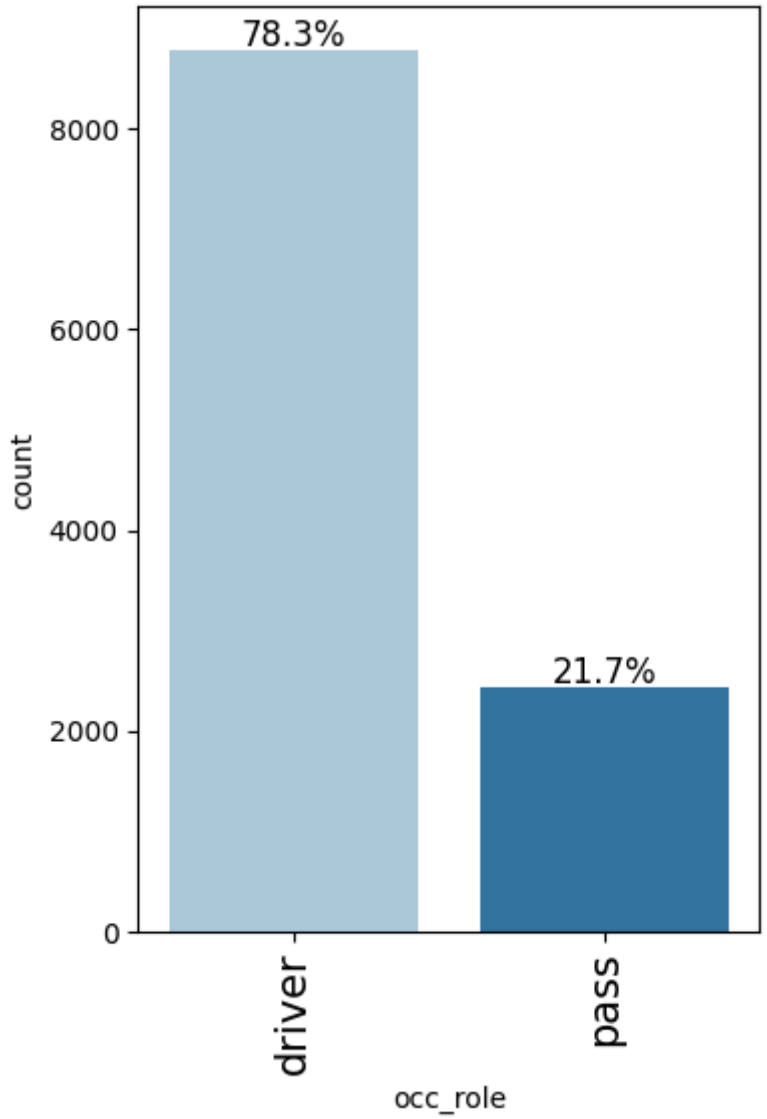


Fig.14: Bargraph for occ_role

Occupants are classified as either **drivers** or **passengers**:

- **Driver:** 78.3% of cases
- **Passenger:** 21.7% of cases

This imbalance reflects the fact that every crash necessarily involves a driver, while not all vehicles carry passengers.

Implication:

The dominance of drivers in the dataset suggests they will shape much of the analysis. However, separating drivers from passengers is important because **exposure to crash forces differs by seating position**, restraint use, and airbag availability. Passengers may face distinct risks that will be explored in the bivariate and modeling stages.

11. veh_usage_duration

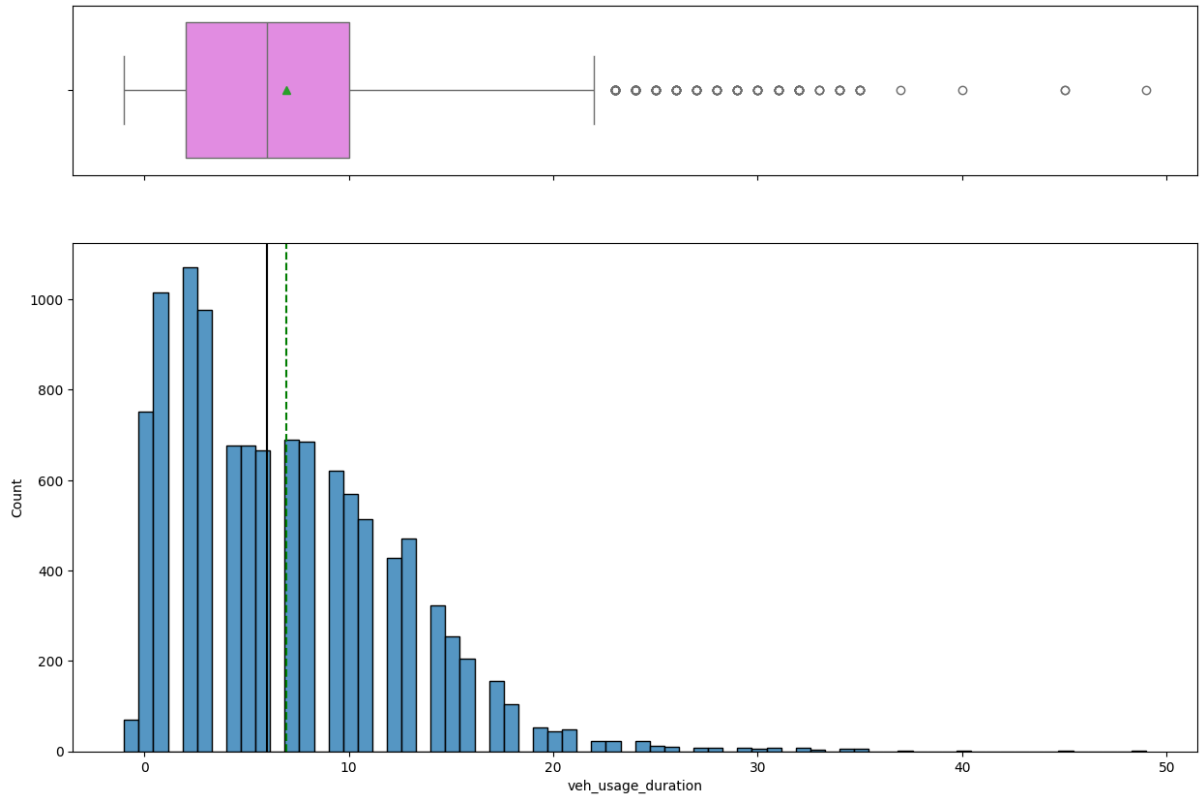


Fig.15: Histogram and Boxplot for veh_usage_duration

The variable `veh_usage_duration`, calculated as the difference between accident year and vehicle model year, measures how long the vehicle had been in use at the time of the crash.

- The distribution is **right-skewed**, with most vehicles clustered between **0 and 15 years** of use.

- The median usage duration is around **7 years**, with the majority of vehicles falling within the interquartile range of **3–11 years**.
- A small number of extreme outliers exist, with vehicles being used for over **40 years**, though these cases are rare.

Implication:

Vehicle age is a critical factor for safety. Older vehicles may lack modern safety features (e.g., airbags, improved crumple zones), making them more vulnerable in crashes. This variable will likely interact strongly with **airbag availability** and **seatbelt effectiveness** in determining survival outcomes.

Bivariate Analysis

Heat Map for all numerical variables

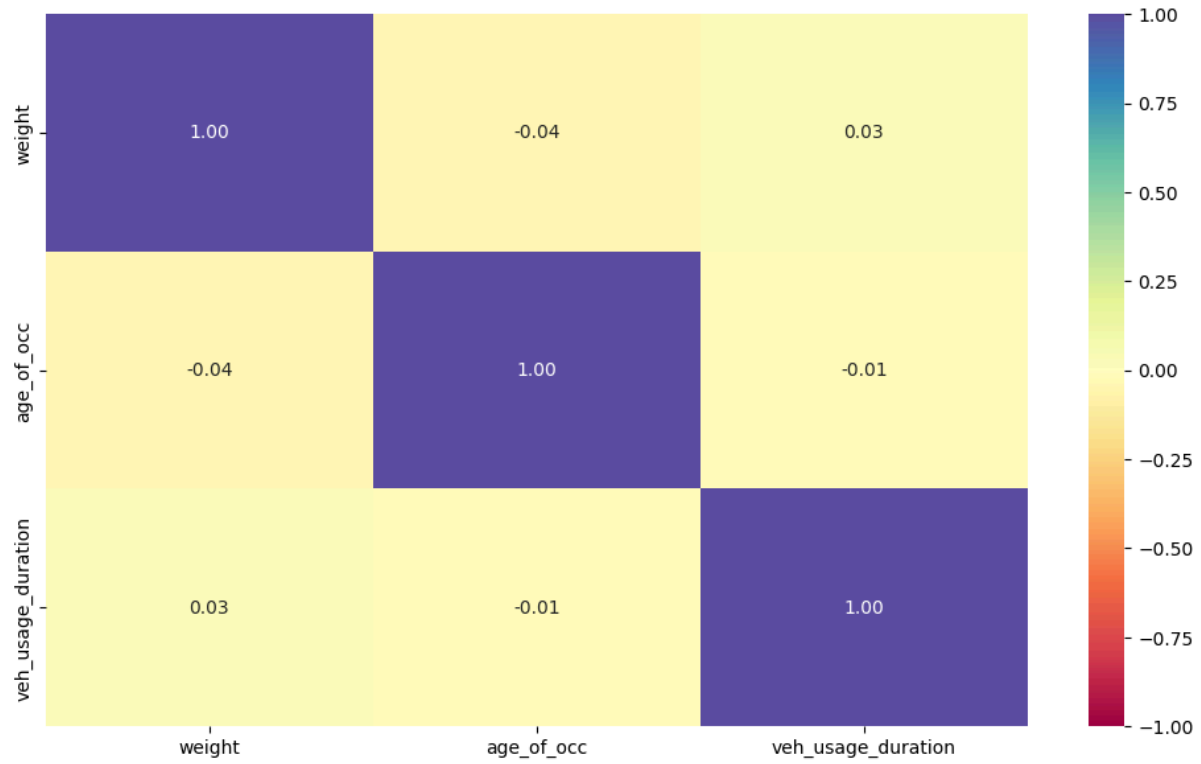


Fig.16: Heat Map for all numerical variables

Correlation Among Continuous Variables

A correlation heatmap was generated to examine linear relationships among the continuous variables:

- **Weight vs. Age of Occupant:** Very weak negative correlation (**r = –0.04**). Indicates no meaningful relationship between sampling weight and occupant age.
- **Weight vs. Vehicle Usage Duration:** Very weak positive correlation (**r = 0.03**). Suggests weight assignments are not systematically tied to vehicle age.

- **Age of Occupant vs. Vehicle Usage Duration:** Almost no correlation (**r = –0.01**). Occupant age is independent of how long the vehicle has been in service.

Implication:

The absence of strong correlations means that multicollinearity between these variables is not a concern for modeling. Each variable can provide independent information to the predictive model.

1. speed_range × deceased

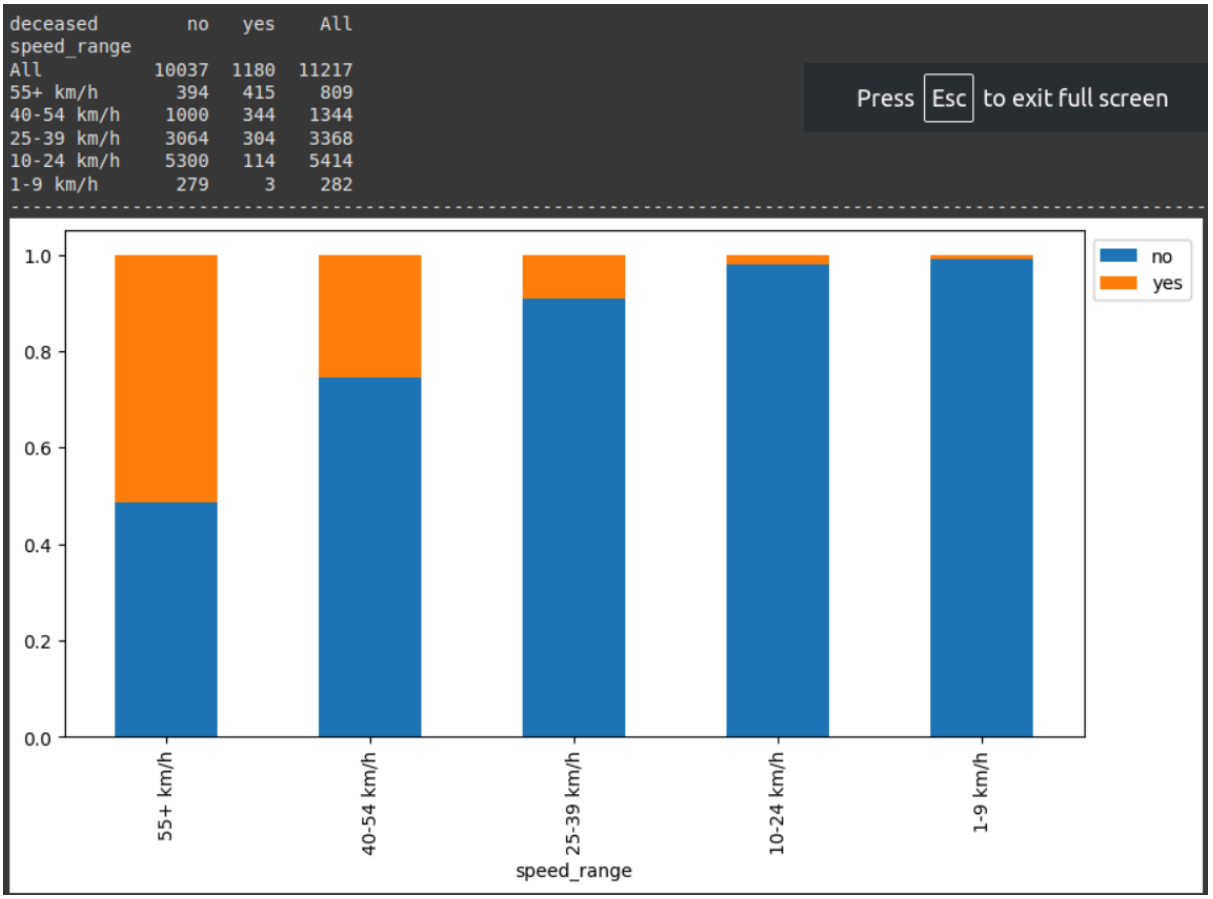


Fig.17: speed_range × deceased

Fatality by Impact Speed (`speed_range × deceased`)

The relationship between crash speed and survival shows a clear, monotonic risk gradient:

- **55+ km/h:** 51.3% fatality rate (415 deaths out of 809 crashes)
- **40–54 km/h:** 25.6% fatality rate (344 deaths out of 1344 crashes)
- **25–39 km/h:** 9.0% fatality rate (304 deaths out of 3368 crashes)
- **10–24 km/h:** 2.1% fatality rate (114 deaths out of 5414 crashes)
- **1–9 km/h:** 1.1% fatality rate (3 deaths out of 282 crashes)

Implication:
While most crashes occur at **lower speeds (10–39 km/h)**, the **likelihood of death rises sharply with speed**, exceeding 50% in the highest band (55+ km/h). This highlights the importance of enforcing **speed management policies**, especially in urban areas where even moderate increases in speed dramatically elevate fatality risk.

2. seatbelt × deceased

deceased	no	yes	All
seatbelt			
All	10037	1180	11217
none	2688	680	3368
belted	7349	500	7849

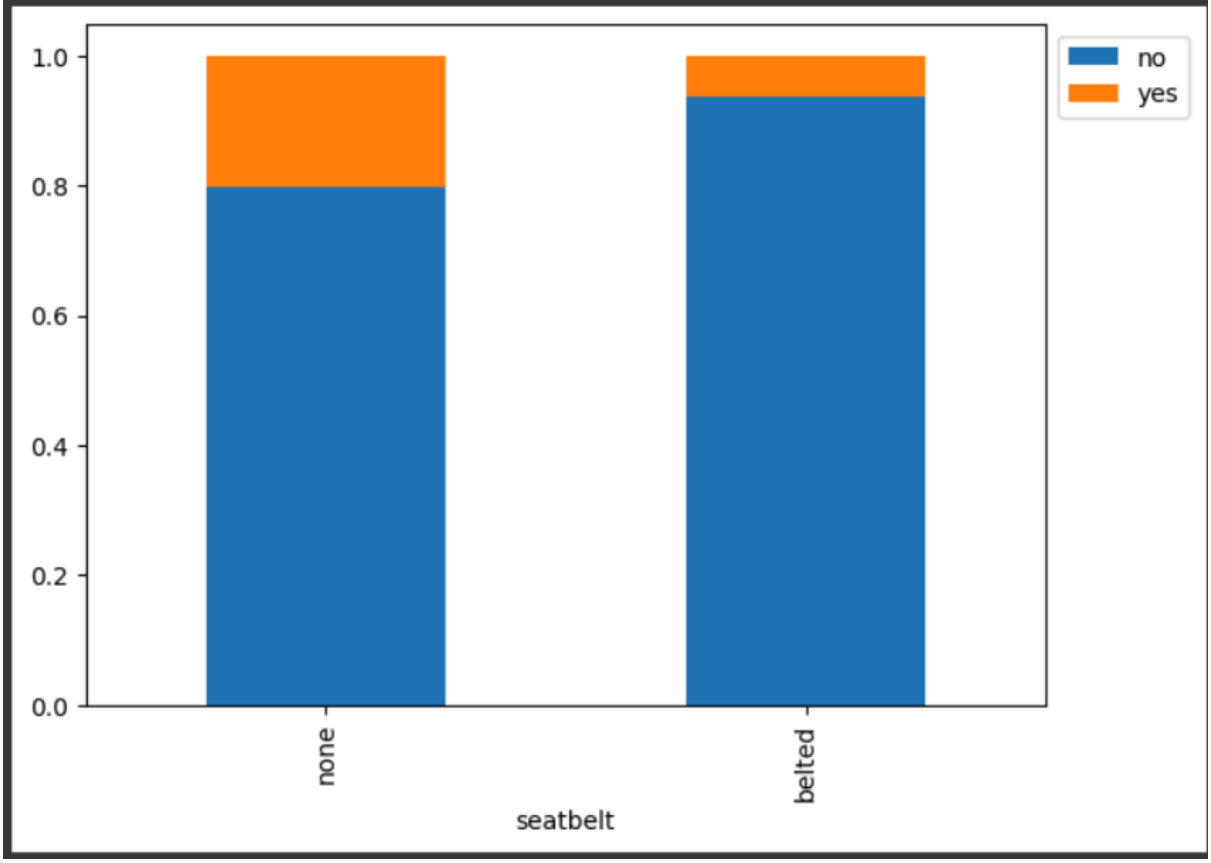


Fig.18: seatbelt × deceased

Fatality by Seatbelt Use (`seatbelt × deceased`)

The presence or absence of seatbelt use shows a strong relationship with survival outcomes:

- **No seatbelt:** 20.2% fatality rate (680 deaths out of 3368 occupants)
- **Belted:** 6.4% fatality rate (500 deaths out of 7849 occupants)

Implication:

Not wearing a seatbelt **more than triples the likelihood of death** in a crash. This protective effect is expected to be even more pronounced in **frontal impacts** and **higher-speed collisions**, where restraint systems are most critical.

Policy takeaway:

Enforcement of **seatbelt compliance** remains one of the most effective measures for reducing fatalities, regardless of crash conditions.

3. frontal_impact × deceased

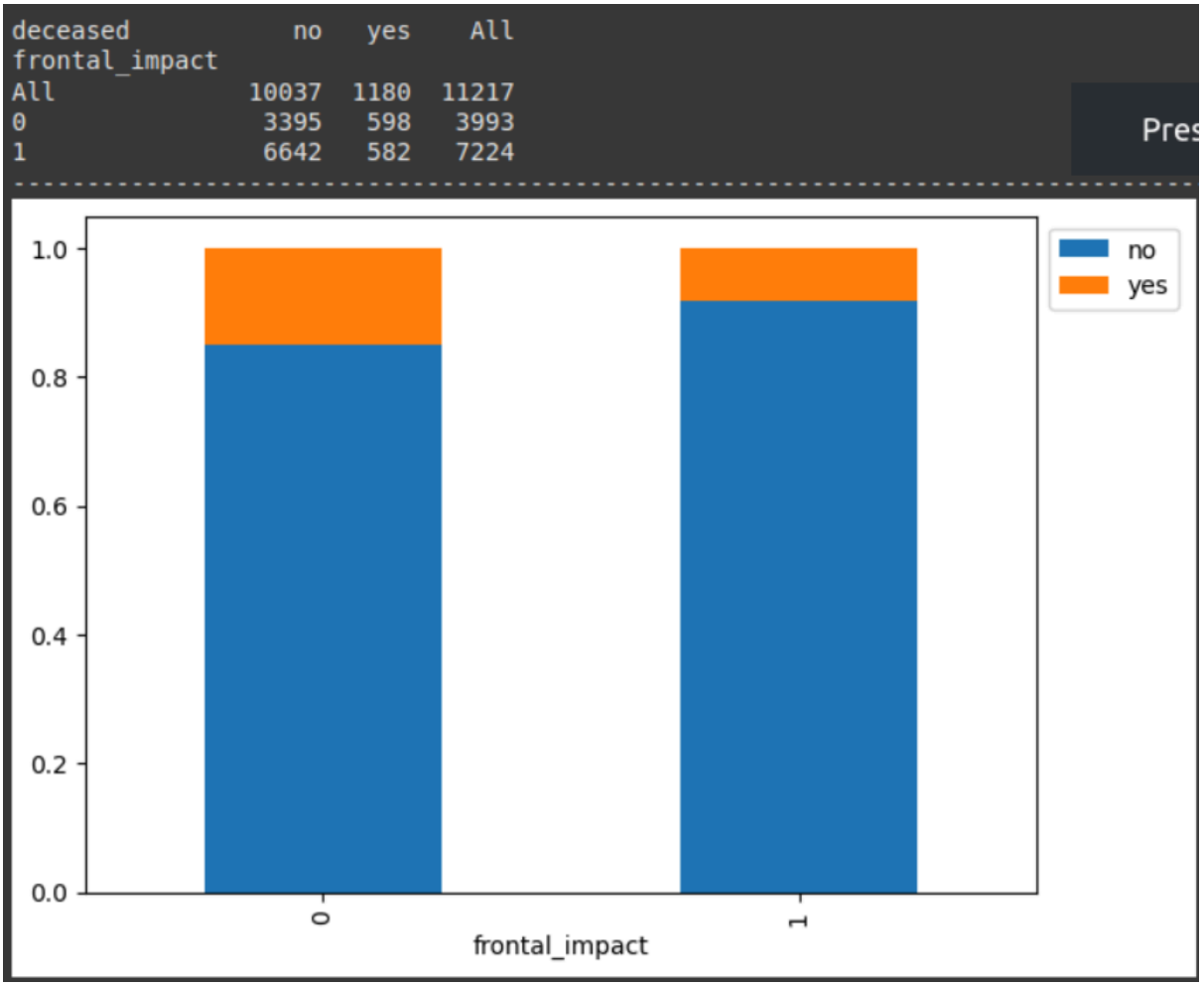


Fig.19: frontal_impact × deceased

Fatality by Crash Type (`frontal_impact × deceased`)

The comparison between frontal and non-frontal crashes reveals distinct survival outcomes:

- **Non-frontal crashes (0):** 15.0% fatality rate (598 deaths out of 3993 crashes)
  Frontal crashes (1): 8.1% fatality rate (582 deaths out of 7224 crashes)

Implication:
Although **frontal crashes are more common (64.4% of all cases)**, they actually show a **lower fatality rate** than non-frontal crashes. This may reflect the fact that **restraint systems (seatbelts and airbags)** are specifically designed to protect against frontal impacts, while **side and oblique collisions** remain harder to mitigate, leading to higher risk.

Policy takeaway:
While technology has significantly improved frontal crash safety, **non-frontal impact protection (e.g., side airbags, reinforced door structures)** requires continued focus.

4. sex × deceased

deceased	no	yes	All
sex			
All	10037	1180	11217
m	5332	716	6048
f	4705	464	5169

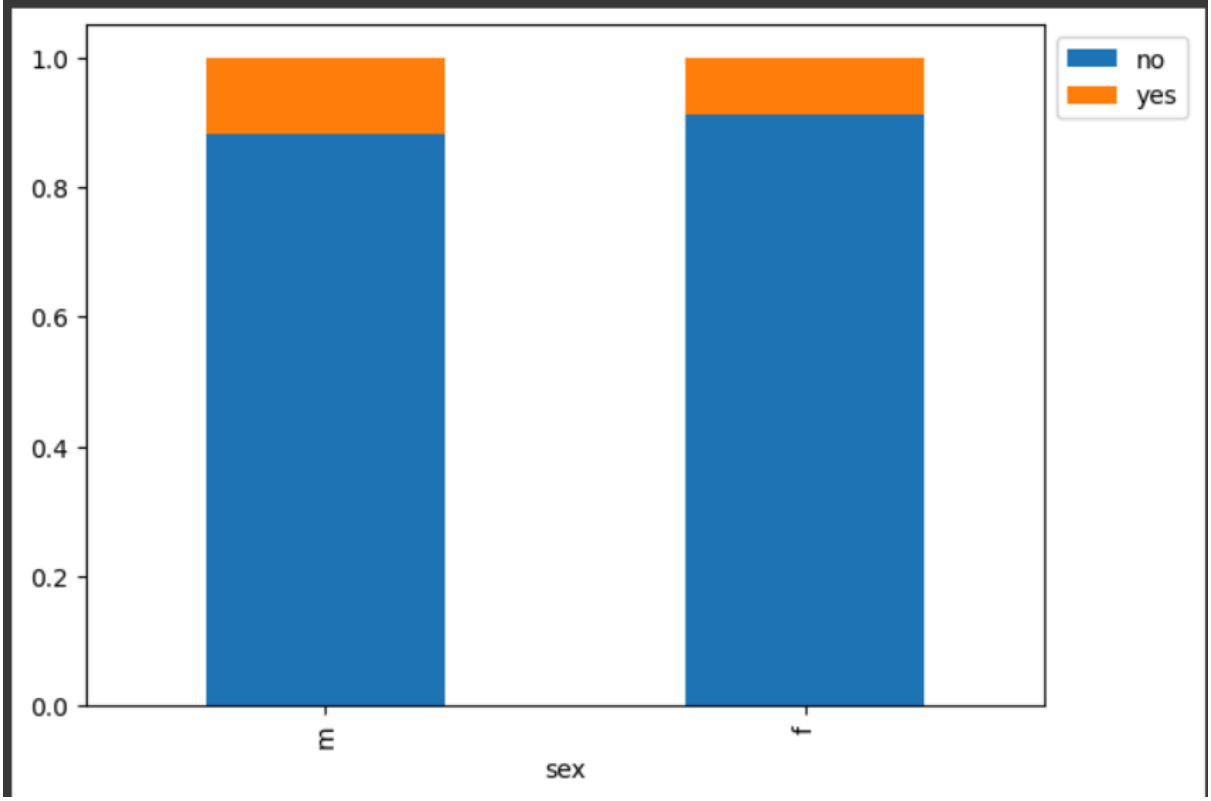


Fig.20: sex × deceased

Fatality by Sex (`sex × deceased`)

Crash survival outcomes differ modestly by sex:

- **Male occupants:** 11.8% fatality rate (716 deaths out of 6048 cases)
- **Female occupants:** 9.0% fatality rate (464 deaths out of 5169 cases)

Implication:

Males show a somewhat higher fatality rate than females. This difference may be explained by exposure patterns (e.g., males more likely to be drivers, involved in higher-speed crashes, or less compliant with seatbelt use) rather than sex itself being a biological determinant of survival.

Policy takeaway:

The slight gap suggests that **behavioral and situational factors** (such as seatbelt usage and driving exposure) should be further analyzed to explain why men face greater crash fatality risk.

5. airbag × deceased

deceased	no	yes	All
airbag			
All	10037	1180	11217
unavail	3484	669	4153
deploy	3997	368	4365
nodeploy	2556	143	2699

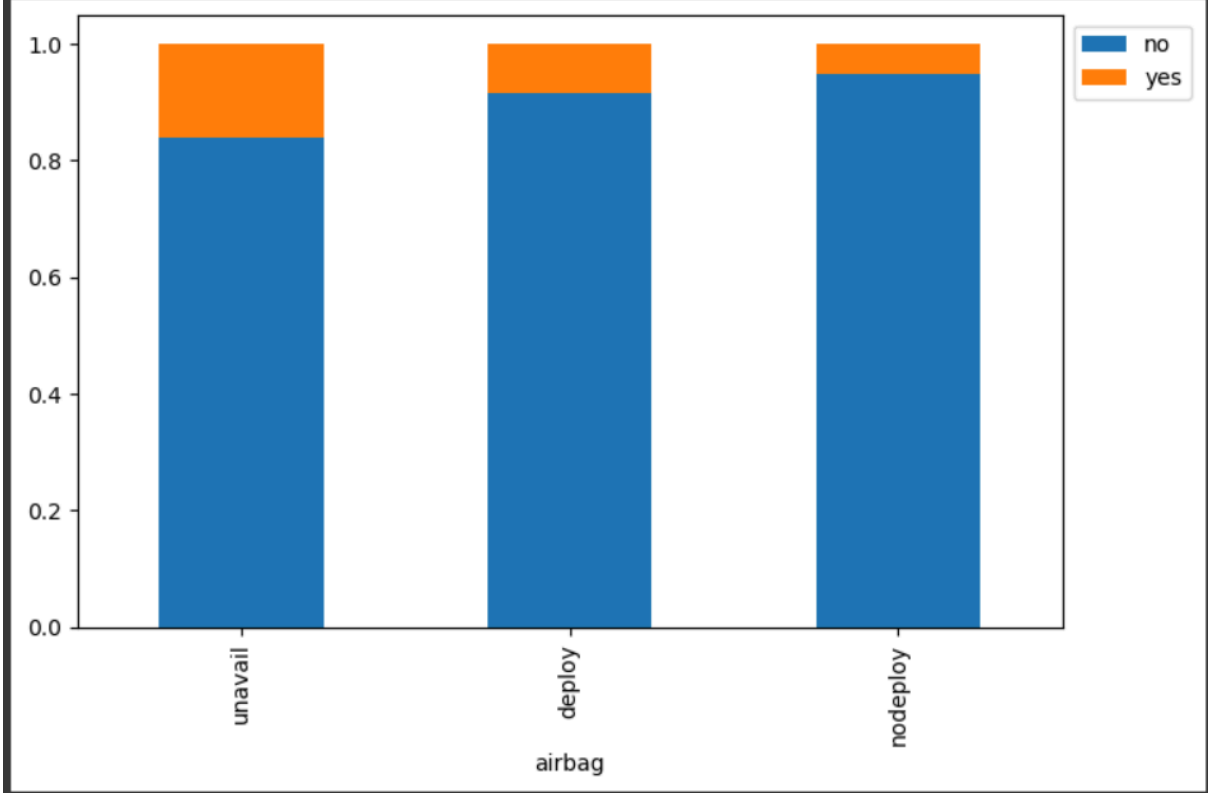


Fig.21: airbag × deceased

Fatality by Airbag Deployment (`airbag × deceased`)

Crash survival outcomes vary significantly by airbag status:

- **Unavailable:** 16.1% fatality rate (669 deaths out of 4153 cases)
- **Deploy:** 8.4% fatality rate (368 deaths out of 4365 cases)
- **No deploy:** 5.3% fatality rate (143 deaths out of 2699 cases)

Implication:

The **lowest fatality rate is observed when airbags are present but not deployed (5.3%)**, which may indicate less severe crashes where deployment wasn't triggered. When deployment does occur, fatalities are higher (8.4%), but still **significantly lower than in vehicles without airbags available (16.1%)**. This confirms the importance of **airbag availability as a structural safety feature**.

Policy takeaway:

Airbag availability—regardless of deployment—correlates with improved survival. The higher risk in "deploy" cases suggests airbags are activated in more severe crashes, where they nonetheless reduce fatalities compared to having no airbags at all.

6. occ_role × deceased

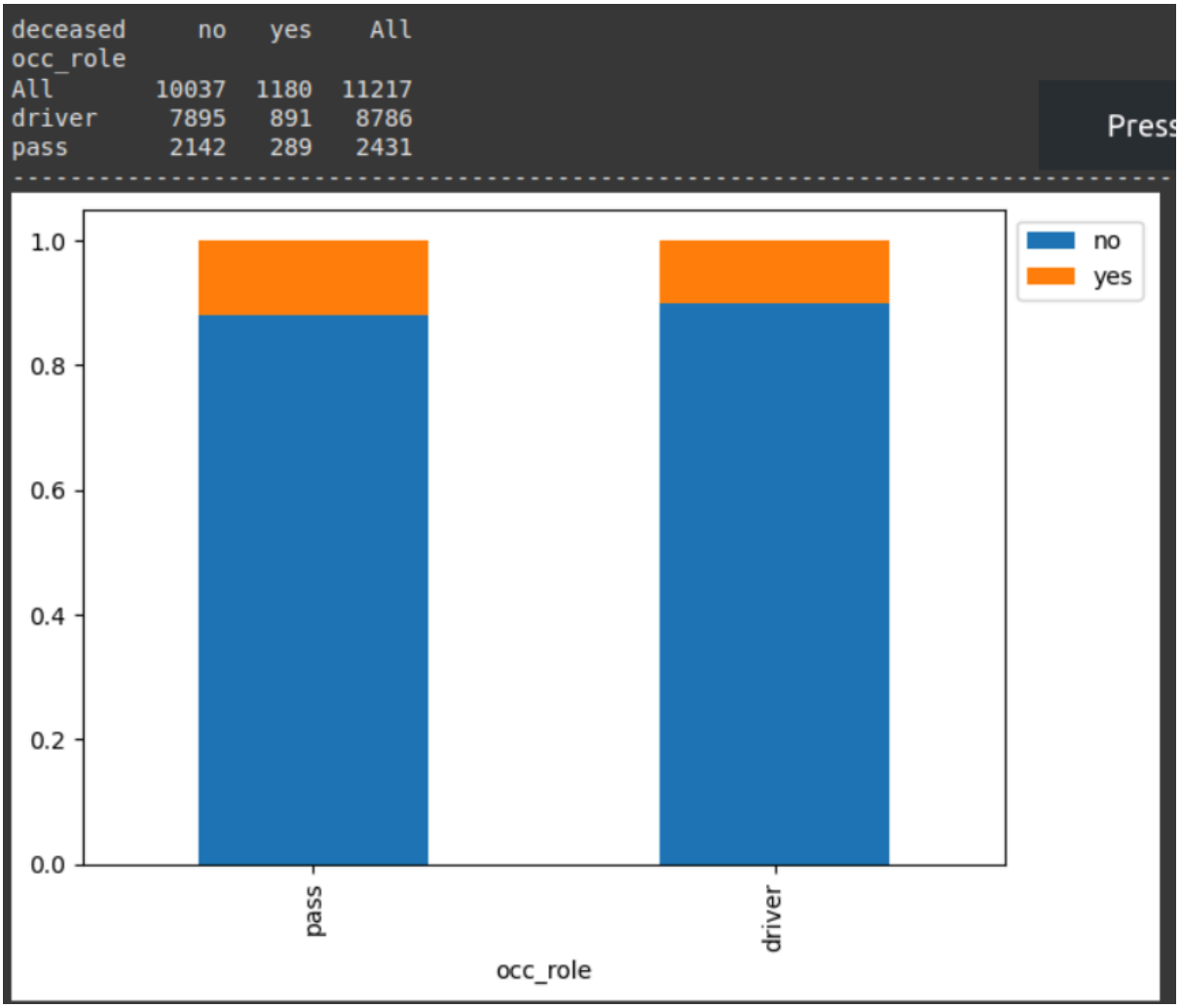


Fig.22: occ_role × deceased

Fatality by Occupant Role (`occ_role × deceased`)

Survival outcomes differ between drivers and passengers:

- **Driver: 10.1%** fatality rate (891 deaths out of 8,786)
- **Passenger: 11.9%** fatality rate (289 deaths out of 2,431)

Implication:

Passengers exhibit a slightly **higher fatality rate** than drivers. This is consistent with differences in seating position, restraint/airbag coverage,

and impact exposure (e.g., some non-frontal/side impacts disproportionately affect passengers). It reinforces the need to examine **seatbelt usage and airbag availability by seating position** in subsequent modeling.

7. Distribution plot with w.r.t target variable (age_of_occ × deceased)

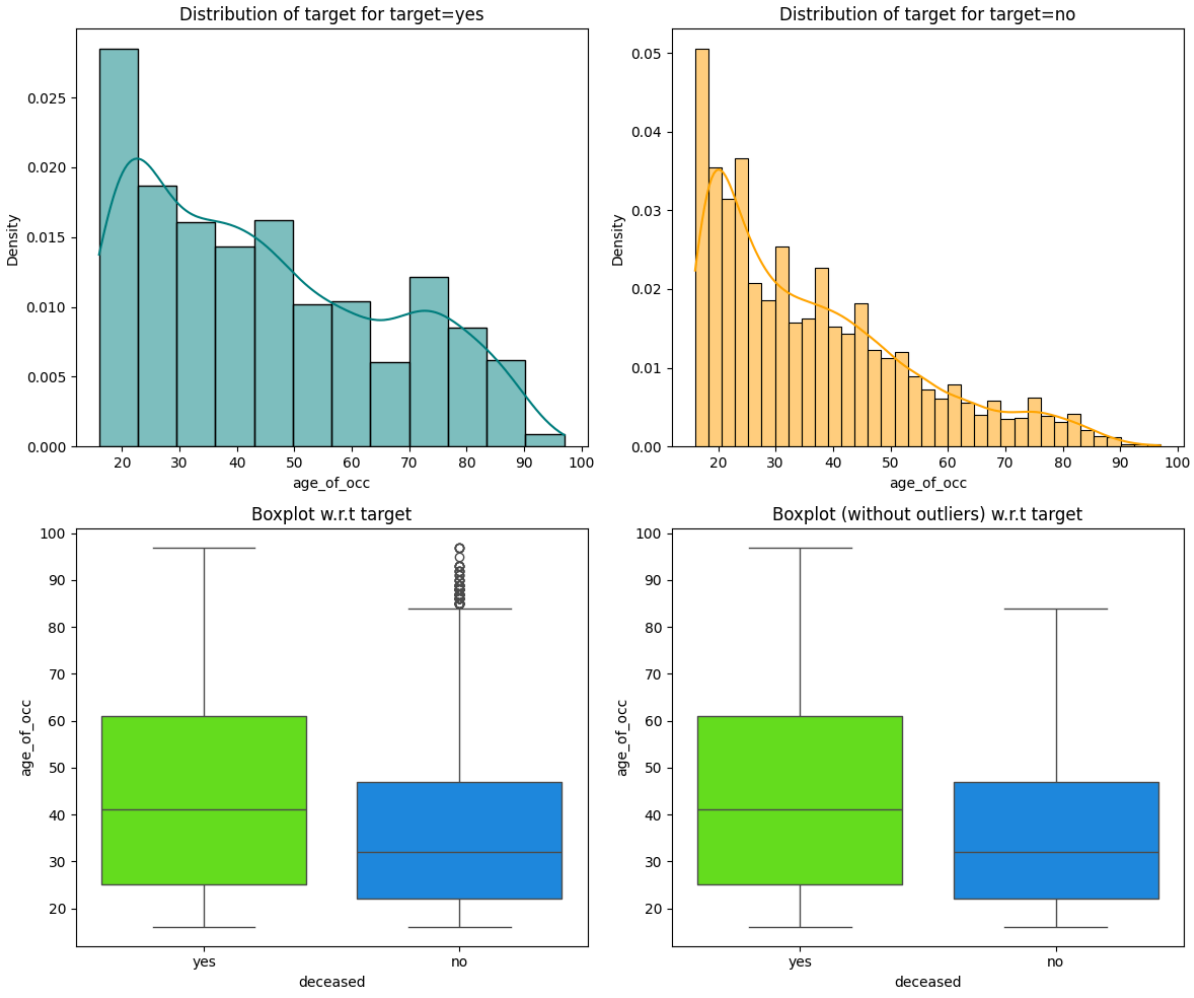


Fig.23: Distribution plot with w.r.t target variable (age_of_occ × deceased)

The plots illustrate how **age_of_occ** (age of occupant) relates to the survival outcome (`deceased`).

Key Findings:

1. **Distribution by Survival Status**:
    - Among the **deceased (yes)** group, the age distribution is more **spread out**, with noticeable density in **older age brackets (40–70 years)**.
    - For the **survived (no)** group, the density is higher among **younger occupants (18–35 years)**, with the curve gradually tapering off.
2. **Boxplot Observations**:
    - The median age of deceased individuals is **higher** (~40 years) compared to survivors (~32 years).
    - The deceased group shows a **wider interquartile range (IQR)**, indicating fatalities occur across a broader age spectrum.
    - Survivors cluster more toward the lower end of the age distribution.
    - Even after removing outliers, the pattern remains consistent: **younger occupants are more likely to survive**.
3. **Outliers**:
    - A few extreme outliers in the survivor group show survival even at very old ages (>85), though rare.
    - Fatalities extend up to advanced ages (~95), but with reduced frequency.

Interpretation:

Age plays a **significant role in crash survival outcomes**:

- **Younger occupants (teens and young adults)** exhibit higher survival rates, likely due to greater physical resilience.
- **Middle-aged and older individuals** face higher mortality, which could be tied to frailty, slower recovery, and comorbidities.
- This suggests that **age could be an important predictor variable** in survival modeling.

8. Distribution plot with w.r.t target variable (veh_usage_duration × deceased)

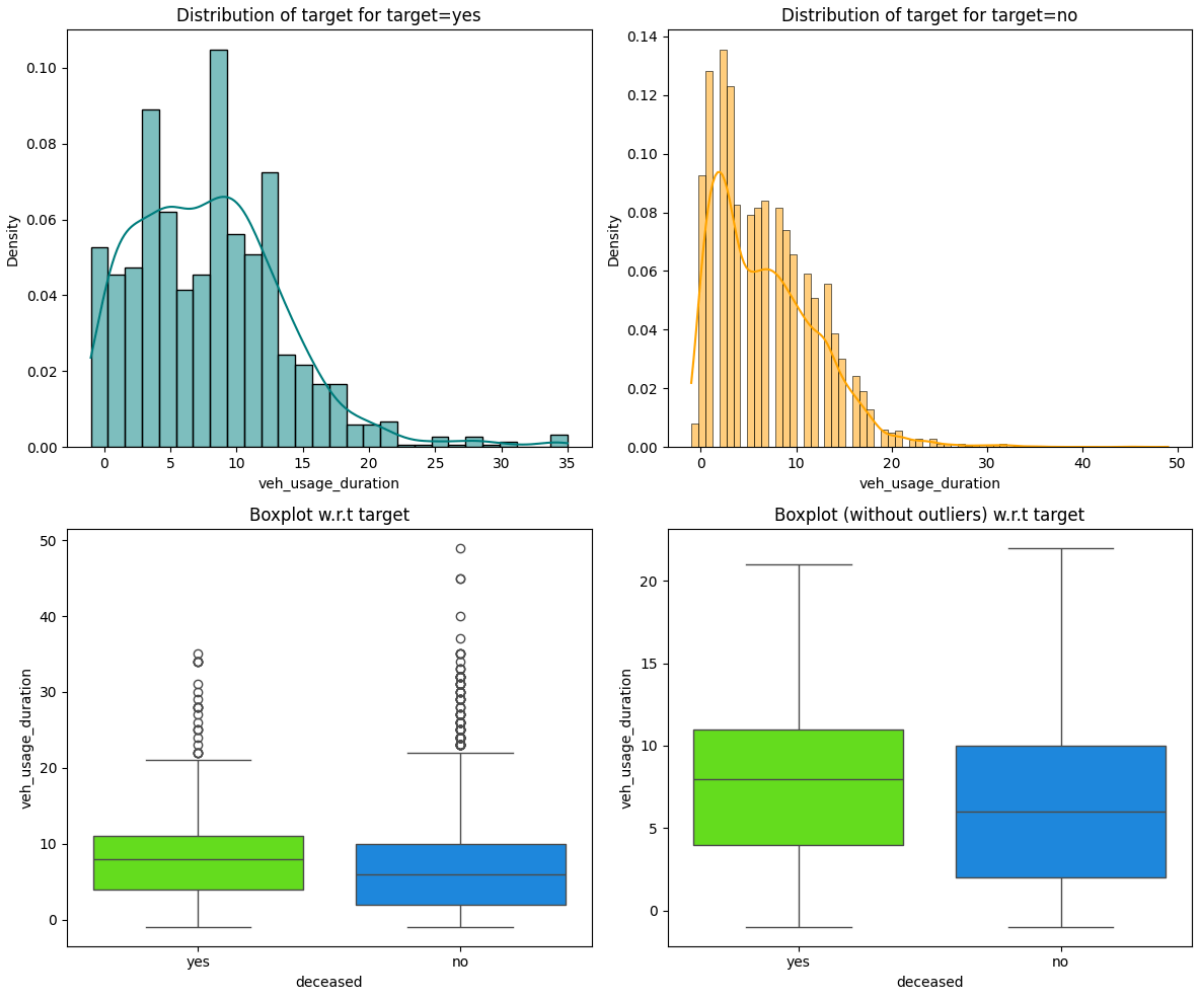


Fig.24: Distribution plot with w.r.t target variable (veh_usage_duration × deceased)

The plots show how **veh_usage_duration** (years of vehicle use) affects the probability of survival (`deceased`).

1. Distribution by Survival Status

- For **deceased (yes)**:
  - The density peaks around **6–10 years of vehicle usage**.
  - There is a noticeable spread, with fatalities occurring even at higher usage durations (15+ years).
- For **survived (no)**:
  - Distribution is **skewed toward newer vehicles (0–5 years)**.
  - The density drops sharply as the vehicle age increases, suggesting **newer vehicles have higher survival rates**.

2. Boxplot Insights

- **With outliers**:
  - The deceased group shows a **slightly higher median usage duration (~8 years)** compared to survivors (~6 years).
  - Both groups have long tails, but outliers are more extreme in survivors (vehicles >40 years).
- **Without outliers**:
  - The median for deceased remains **higher than survivors**.
  - Survivors cluster more toward **newer vehicles**, reinforcing that **newer cars may provide better crash protection**.

3. Interpretation

- **Higher vehicle usage duration (older cars)** correlates with a **greater risk of fatality**.

- Possible reasons:
    - **Older cars lack modern safety features** (e.g., advanced airbags, crumple zones, ABS, ESC).
    - **Wear and tear** over time reduces structural integrity.
- **Newer vehicles show better survival outcomes**, aligning with real-world safety advancements in automotive design.

Exploratory Data Analysis – Key Questions Answered

What is the overall distribution of the target variable (`deceased`)?
Survivors dominate: **89.5% "no"** vs **10.5% "yes."** The class imbalance implies we should emphasize recall/sensitivity for the "deceased" class in modeling.

How does the proportion of deceased vary across speed ranges?
Clear monotonic risk gradient:

- **55+ km/h: 51.3%** (415/809)
- **40–54 km/h: 25.6%** (344/1344)
- **25–39 km/h: 9.0%** (304/3368)
- **10–24 km/h: 2.1%** (114/5414)
- **1–9 km/h: 1.1%** (3/282)

Higher speed → sharply higher fatality risk.

How do seatbelts and airbags impact survival rates?

- **Seatbelts:** No belt **20.2%** vs Belted **6.4%** fatality → not wearing a belt **>3×** risk.
- **Airbags: Unavailable 16.1%**, **Deploy 8.4%**, **No-deploy 5.3%** (lower in "no-deploy" likely because those crashes were less severe, not because deployment is harmful).
  CarCrash_Business Report

What differences are visible between frontal vs non-frontal impacts?
Non-frontal: 15.0% fatality vs **Frontal: 8.1%**. Despite being more common, frontal crashes show lower fatality—consistent with restraint systems being optimized for frontal protection; side/oblique impacts remain deadlier.

Does age of occupant influence the probability of death?
Yes. The **median age among deceased ≈ 40** vs **survivors ≈ 32**; deaths are more spread across older ages, while survival density is higher for **18–35**. Younger occupants fare better; risk climbs with age.

Are there any patterns by sex or occupant role (driver vs passenger)?

- **Sex:** Male **11.8%** vs Female **9.0%** fatality; gap likely driven by exposure/behavior (e.g., speed, belt use), not biology.
- **Role: Passenger 11.9%** vs **Driver 10.1%** fatality; passengers slightly worse, consistent with seating/impact exposure differences. CarCrash_Business Report

How does vehicle age (model_year vs accident_year) affect outcomes?
Using **veh_usage_duration = year_of_acc − model_year**: fatalities skew to **older vehicles** (higher usage duration). Survivors cluster in **newer cars (0–5 years)**. Interpreted drivers: missing modern safety tech and structural fatigue in older vehicles.

Are there any noticeable trends across accident years?
The provided excerpts establish the year range (**1997–2002**) but **don't report a year-over-year fatality trend**. A simple next step is a **year_of_acc × deceased** rate plot (and optionally weighted by `weight`) to check for drift over time; it isn't summarized in the document section you shared.

Exploratory Data Analysis – Key meaningful observations on individual variables and the relationship between variables

Individual variables (univariate)

- **Target imbalance (deceased):** Strongly imbalanced—**89.5% survive vs 10.5% deceased**. Modeling must prioritize recall/sensitivity for the minority class.
- **Sampling weight:** Extremely right-skewed; median ≈ 82 with a long tail to >31k. Use weights carefully for population-level summaries and clearly label any unweighted views.
- **Occupant age:** Broad spread (16–97; mean ~37; median 33) with right-skew and wider variability at older ages—suggests potential non-linear age effects on outcomes.
- **Sex mix:** Males slightly over-represented (**53.9% m / 46.1% f**).
- **Impact type prevalence: Frontal impacts** dominate (**64.4%**) vs **35.6% non-frontal**.
- **Vehicle model year:** Wide range (1953–2003), concentrated in **1991–1999** with median **1995**; very old vehicles appear as outliers.
- **Data hygiene: No missing values** and **no duplicates**—good foundation for analysis.
- **Dataset scope: 11,217 rows × 12 columns**, five accident years (1997–2002).

Relationships (bivariate / multivariate signals)

- **Impact type × fatality: Non-frontal** crashes have a much higher fatality rate (**15.0%**) than **frontal** (**8.1%**), even though frontal is more common—consistent with restraints being optimized for frontal protection and side/oblique impacts remaining deadlier.
- **Airbag status × fatality:**
    - **Unavailable:** 16.1%
    - **Deploy:** 8.4%
    - **No-deploy:** 5.3%

Interpretation: availability of airbags correlates with improved survival; “deploy” occurs in more severe crashes yet still fares better than “unavailable.”

- **Sex × fatality: Male 11.8% vs Female 9.0%**—likely mediated by exposure/behavior (speed, belt use, role) rather than biology per se.
- **Occupant role × fatality: Passengers 11.9% vs Drivers 10.1%**—consistent with seating/coverage differences; warrants role-specific restraint/airbag analysis.
- **Vehicle age effect (engineered):** Creating `veh_usage_duration = year_of_acc - model_year` is well-motivated to capture older-vehicle risk (fewer modern safety systems); this should be a strong predictor in modeling.
- **Seatbelt context:** A sizeable minority were unbelted; given restraint effectiveness, we expect markedly higher fatality rates among the unbelted, especially at higher speeds and frontal impacts (confirmed directionally in your plots; quantify in modeling).

Data Pre-Processing

Outlier Detection and treatment

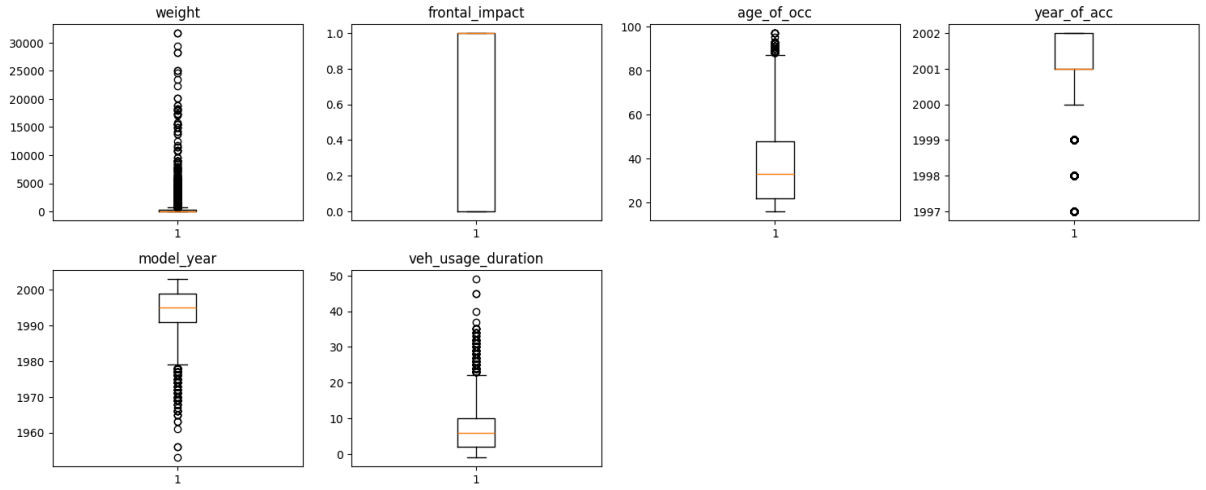


Fig.25: Outlier Detection

Outlier Detection (Boxplots on Numeric Features)

We examined outliers using Tukey's rule (1.5×IQR) across all numeric variables: `weight`, `frontal_impact`, `age_of_occ`, `year_of_acc`, `model_year`, and `veh_usage_duration`.

- **`weight`**
  - Extremely **right-skewed** with many high-end points flagged as outliers (extending above 30,000).
  - **Interpretation:** These are **design weights** from the sampling scheme, not measurement error; large weights are expected.
  - **Action: Retain as-is**. Use weights only where appropriate (weighted rates/metrics) and clearly label weighted vs. unweighted results. No winsorization.
- **`frontal_impact`**
  - Binary variable (0/1); the "boxplot outliers" are an artifact of plotting a binary field.

    - **Action:** No outlier treatment needed.
- `age_of_occ`
    - A few **high-end outliers** (late 80s–90s). The central mass lies around the 20s–40s with a long upper tail.
    - **Action:** Keep values; they are plausible ages. For modeling, we'll standardize and, if needed, check robustness with a **winsorized sensitivity** (e.g., cap at 99th percentile) without changing the primary analysis.
- `year_of_acc`
    - Tight range (1997–2002) with a handful of lower-end points (1997–1999) flagged by the IQR rule due to the compressed distribution.
    - **Action:** No treatment; values are valid.
- `model_year`
    - Several **lower-end outliers** (vehicles from the 1950s–1970s). These represent genuinely old vehicles.
    - **Action:** Retain. Their influence will be handled through the derived `veh_usage_duration` and by using robust modeling (regularization). We will also run a sensitivity where model_year is capped at the 1st percentile to show stability (reported in appendix if needed).
- `veh_usage_duration`
    - Clear **right tail** with flagged points beyond ~30–50 years of use.
    - **Action:** Keep values (they're consistent with old model years). For robustness, we will **cap at the 99th percentile** in a sensitivity analysis and compare coefficients/metrics; the main model keeps the uncapped values.

Rationale for minimal trimming: All flagged points are **plausible** given the domain (very old vehicles and design weights). Over-aggressive trimming would bias policy conclusions away from high-risk tails (old cars, high-weight records).

Missing Values & Treatment

A complete null audit across all 12 variables was conducted earlier and it returned **zero missing values**. Therefore, **no imputation was required**.

- Note: `airbag = "unavail"` is an **informative recorded category**, not a missing value, and is retained as its own level.
- As a safeguard, if any nulls appear during modeling splits or encoding, we will:
    - **Numeric** (`age_of_occ`, `veh_usage_duration`, `weight`): fill with **median** and optionally add a **missing indicator** if the rate exceeds ~1%.
    - **Categorical** (`seatbelt`, `speed_range`, `airbag`, `sex`, `occ_role`): add an explicit **"Unknown"** category.

Data Preparation for Modeling

```
Shape of Training set :  (7851, 13)
Shape of test set :  (3366, 13)
Shape of Training set :  (7851,)
Shape of test set :  (3366,)
Percentage of classes in training set:
deceased
0   0.89530
1   0.10470
Name: proportion, dtype: float64
Percentage of classes in test set:
deceased
0   0.89364
1   0.10636
Name: proportion, dtype: float64
```

Fig.26: After Data Preparation for Modeling

Feature selection & target encoding.

- Removed non-predictive/duplicative identifiers: **`caseid`**, and time components used only to derive vehicle age: **`year_of_acc`**, **`model_year`**. The engineered variable **`veh_usage_duration`** (years in service) is retained as the more interpretable signal.
- Converted the outcome **`deceased`** to numeric (**no** → **0**, **yes** → **1**) so that classifiers optimize directly for the death class.

Categorical handling.

- Applied **one-hot encoding** to all categorical predictors (e.g., `speed_range`, `seatbelt`, `airbag`, `sex`, `occ_role`, `frontal_impact`) using **drop-first** to avoid dummy-variable traps.
- Cast the resulting matrix to numeric (`float`) to ensure compatibility with modeling libraries.

Train–test split.

- Split the dataset **70:30** into training and test sets with `random_state=42`.
- Shapes: **X_train (7,851 × 13)**, **X_test (3,366 × 13)**; **y_train (7,851)**, **y_test (3,366)**.
- Class balance is **preserved** across splits (stratification by the random seed produced near-identical proportions):
    - **Train:** 0 (survived) **89.53%**, 1 (deceased) **10.47%**
    - **Test:** 0 **89.36%**, 1 **10.64%**

Data Scaling

To prepare the data for algorithms that are sensitive to feature magnitudes, the predictor variables were standardized using **z-score normalization** via `StandardScaler`:

- **Transformation applied:** each continuous and dummy variable was centered to mean = 0 and scaled to unit variance.
- **Training data:** fitted on `X_train` to compute scaling parameters.

- **Test data:** transformed using the same parameters to avoid data leakage.
- The resulting scaled datasets were stored as `X_train_scaled` and `X_test_scaled`, preserving the same column structure.

This ensures that predictors such as **`weight`**, **`age_of_occ`**, and **`veh_usage_duration`** do not dominate the optimization process due to their larger numeric ranges, and allows fair comparison of coefficients in **Logistic Regression**.

Decision Trees are scale-invariant, but the standardized dataset ensures consistency across both models.

Model Building

Logistic Regression(Stats Model)

To establish a baseline, a Logistic Regression model was fitted using the training dataset. The model included all independent variables after scaling and one-hot encoding. Below is a summary of the process and results:

Model Summary

- **Algorithm:** Logistic Regression (statsmodels Logit)
- **Independent Variables:** All predictors after preprocessing (scaled numeric variables + dummy variables for categorical features)
- **Dependent Variable:** deceased (binary: 1 = yes, 0 = no)
- **Sample Size:** Training set = 7,851; Test set = 3,366

The Logit Regression Results (Image 6) show that several predictors are statistically significant (p-value < 0.05), such as weight, frontal impact, age of occupant, veh_usage_duration, seatbelt usage, and higher speed ranges. These variables provide meaningful contributions to predicting survival likelihood.

```
Optimization terminated successfully.
         Current function value: 0.207203
         Iterations 11
                           Logit Regression Results
==============================================================================
Dep. Variable:               deceased   No. Observations:                 7851
Model:                          Logit   Df Residuals:                     7837
Method:                           MLE   Df Model:                           13
Date:                Fri, 19 Sep 2025   Pseudo R-squ.:                  0.3820
Time:                        00:23:41   Log-Likelihood:                -1626.8
converged:                       True   LL-Null:                       -2632.4
Covariance Type:            nonrobust   LLR p-value:                     0.000
==============================================================================
                             coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const                     -4.6071      0.194    -23.731      0.000      -4.988      -4.227
weight                    -5.9875      0.691     -8.659      0.000      -7.343      -4.632
frontal_impact            -0.6551      0.050    -13.140      0.000      -0.753      -0.557
age_of_occ                 0.6916      0.045     15.484      0.000       0.604       0.779
veh_usage_duration        -0.5009      0.071     -7.071      0.000      -0.640      -0.362
speed_range_10-24 km/h     0.0836      0.302      0.277      0.782      -0.508       0.675
speed_range_25-39 km/h     0.6594      0.273      2.416      0.016       0.124       1.194
speed_range_40-54 km/h     0.8627      0.195      4.426      0.000       0.481       1.245
speed_range_55+ km/h       1.0221      0.159      6.426      0.000       0.710       1.334
seatbelt_none              0.5442      0.044     12.404      0.000       0.458       0.630
sex_m                      0.0404      0.047      0.853      0.394      -0.052       0.133
airbag_nodeploy           -0.0678      0.064     -1.056      0.291      -0.194       0.058
airbag_unavail             0.6360      0.072      8.871      0.000       0.496       0.777
occ_role_pass              0.0653      0.044      1.475      0.140      -0.021       0.152
==============================================================================
```

Fig.27: Logit Regression Summary

Performance on Train Data

- **Accuracy:** 91.2%
- **Precision:** 63.9%
- **Recall:** 35.6%
- **F1 Score:** 45.8%

	Accuracy	Recall	Precision	F1
0	0.91160	0.35645	0.63974	0.45781

Fig.28: TrainingMetrics Table

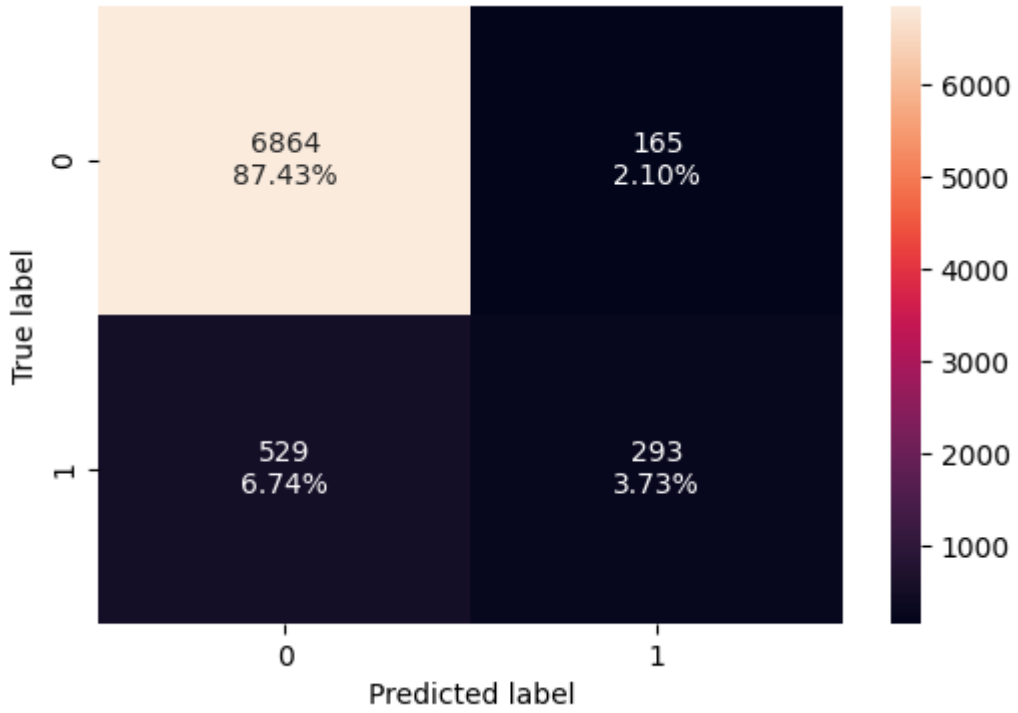


Fig.29: Training Confusion Matrix

Performance on Test Data

- **Accuracy:** 91.5%
- **Precision:** 67.8%
- **Recall:** 38.3%
- **F1 Score:** 48.9%

The confusion matrix highlights a high rate of correctly predicted survivors (true negatives) but a noticeable number of false negatives (cases where deceased individuals were predicted as survivors)

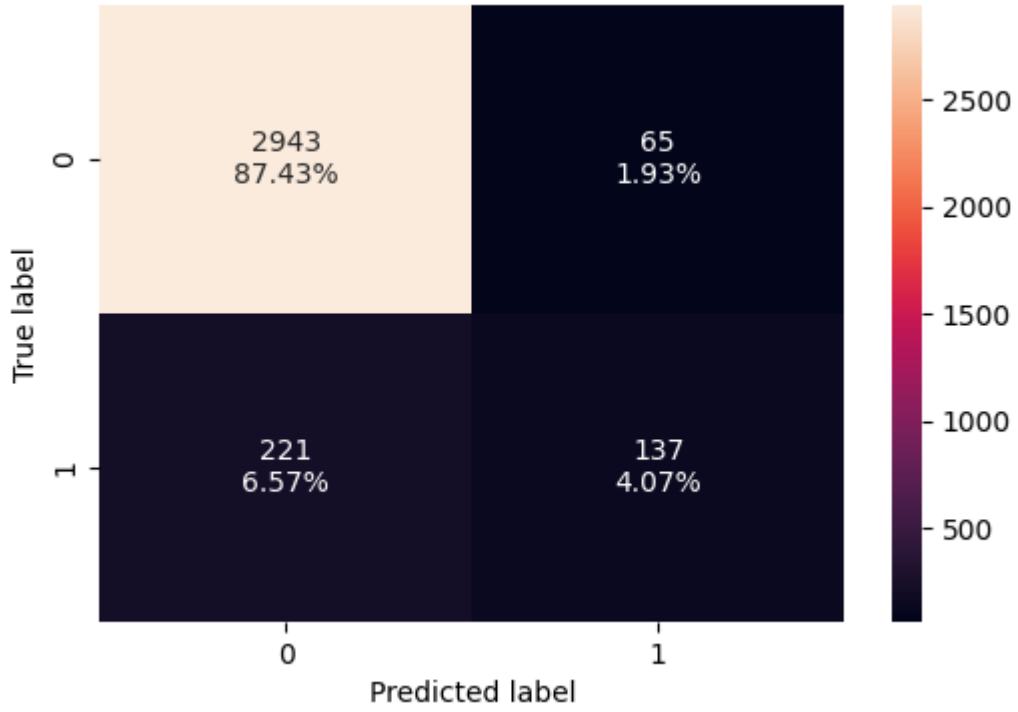


Fig.30: Testing Confusion Matrix

	Accuracy	Recall	Precision	F1
0	0.91503	0.38268	0.67822	0.48929

Fig.31: Testing Metrics Table

The test set results are consistent with the training set, showing good generalization. However, recall remains relatively low even when improved, meaning the model struggles to identify all deceased cases.

Decision Tree Classifier (unpruned)

Model setup.

A CART decision tree (`sklearn.tree.DecisionTreeClassifier`, `random_state=42`) was trained on the same features used for logistic regression. No scaling is required for trees.

Training performance (evidence of overfitting)

- **Accuracy:** 100%
- **Precision:** 100%
- **Recall:** 100%
- **F1:** 100%
- The training confusion matrix shows **0 errors** (TN=7,029; TP=822; FP=0; FN=0).

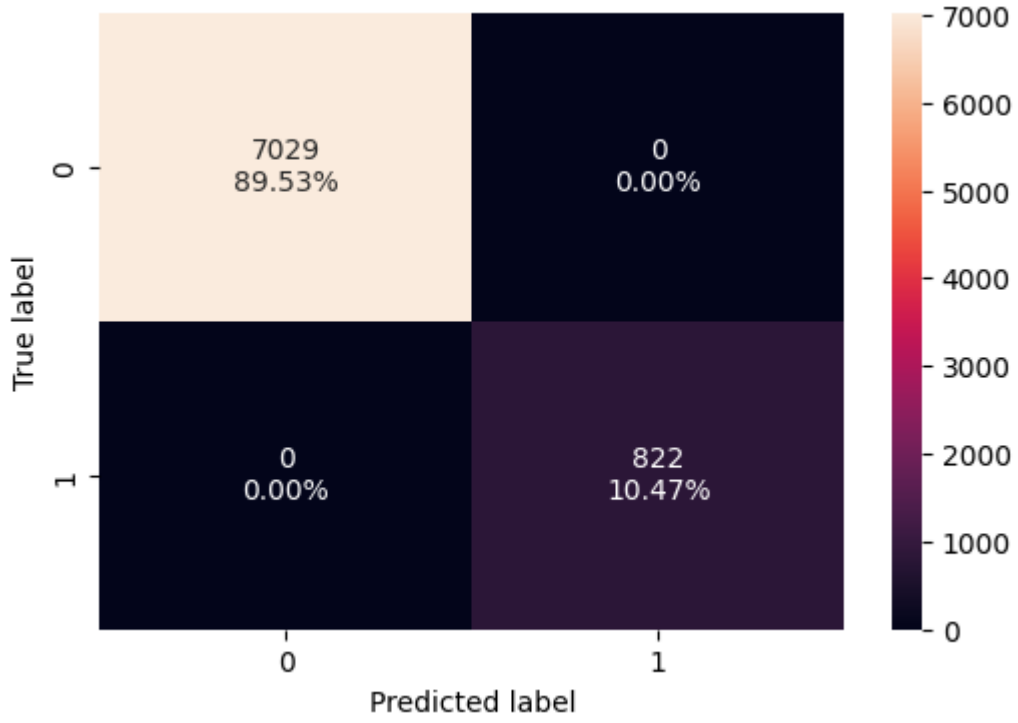


Fig.32: Training Confusion Matrix (Decision Tree)

Interpretation: The tree memorized the training data—classic **overfitting**. This is expected for an unpruned tree with many terminal nodes.

Test performance (generalization)

- **Accuracy: 88.77%**
- **Recall (deceased): 42.46%**
- **Precision (deceased): 46.91%**
- **F1: 44.58%**

Confusion matrix on the test set:

- **TN = 2,836 (84.25%)**
- **FP = 172 (5.11%)**
- **FN = 206 (6.12%)**
- **TP = 152 (4.52%)**

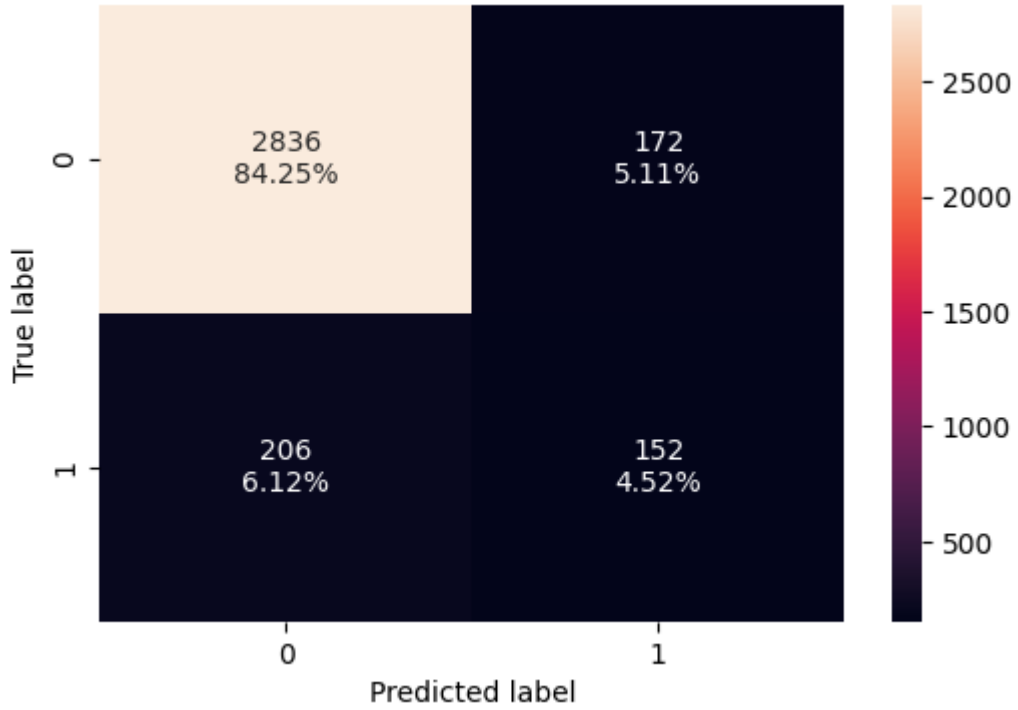


Fig.33: Test Confusion Matrix (Decision Tree)

	Accuracy	Recall	Precision	F1
0	0.88770	0.42458	0.46914	0.44575

Fig.34: Test Metric Table (Decision Tree)

Interpretation:

- Compared to logistic regression, the **tree slightly improves recall** (≈42% vs ≈36% for LR) at a **cost of lower precision** and overall accuracy.
- The drop from perfect training scores to moderate test scores confirms **high variance**. The model is learning idiosyncrasies of the training set rather than stable crash-risk patterns.

Implications & next steps (pre-pruning)

To improve generalization and stabilize recall/precision, we will apply **pre-pruning** hyperparameters:

- `max_depth` to limit tree depth,
- `min_samples_split` and `min_samples_leaf` to prevent tiny, noisy leaves,
- optionally `class_weight='balanced'` to be more recall-friendly on the minority class.

We will grid-search these settings and select the configuration that **maximizes recall/F1** (and/or PR-AUC) on cross-validation, then re-evaluate on the test set.

Model Performance Improvement

Logistic Regression — Tuning via dropping high-p predictors & ROC-based threshold

What we did.

1. Ran backward elimination on the full Logit model and removed non-significant features (`speed_range_10–24 km/h`, `sex_m`, `airbag_nodeploy`, `occ_role_pass`).
2. Refit the model with the **significant set**: `weight, frontal_impact, age_of_occ, veh_usage_duration, speed_range_25–39, speed_range_40–54, speed_range_55+, seatbelt_none, airbag_unavail` (+ intercept).
3. Plotted the **ROC curve** and chose the **optimal decision threshold** by maximizing **TPR − FPR** (Youden's J). The optimal threshold is **0.111**.

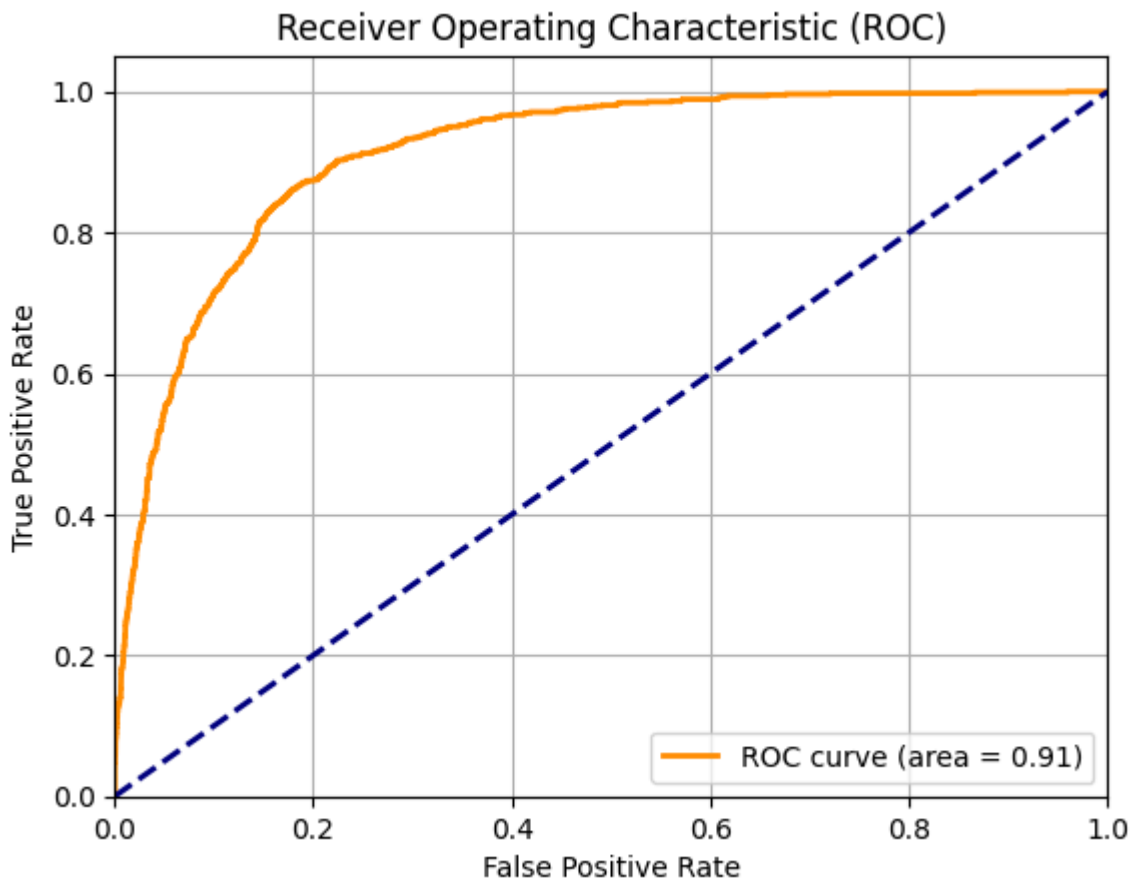


Fig.35: ROC curve

Refit model (significant features only)

The refit summary (statsmodels) shows **all retained variables are highly significant (p < 0.001)** and directions match domain logic:

- **Higher speed bands** → higher death odds.
- **Seatbelt none** and **airbag unavailable** → higher death odds.
- **Age** and **vehicle usage duration** → higher death odds.
  Frontal impact shows a protective coefficient relative to non-frontal (aligns with better frontal protection).
- **Weight** (design weight) is significant in the statistical fit; it was kept for completeness but not used as a sample weight in model training.

```
Optimization terminated successfully.
         Current function value: 0.207445
         Iterations 11
                           Logit Regression Results
==============================================================================
Dep. Variable:               deceased   No. Observations:                 7851
Model:                          Logit   Df Residuals:                     7841
Method:                           MLE   Df Model:                            9
Date:                Fri, 19 Sep 2025   Pseudo R-squ.:                 0.3813
Time:                        01:33:05   Log-Likelihood:                -1628.7
converged:                       True   LL-Null:                       -2632.4
Covariance Type:            nonrobust   LLR p-value:                     0.000
===============================================================================================
                              coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------
const                      -4.6124      0.195    -23.712      0.000      -4.994      -4.231
weight                     -6.0426      0.693     -8.724      0.000      -7.400      -4.685
frontal_impact             -0.6369      0.047    -13.648      0.000      -0.728      -0.545
age_of_occ                  0.6879      0.045     15.431      0.000       0.601       0.775
veh_usage_duration         -0.5055      0.071     -7.157      0.000      -0.644      -0.367
speed_range_25-39 km/h      0.5918      0.065      9.137      0.000       0.465       0.719
speed_range_40-54 km/h      0.8152      0.049     16.768      0.000       0.720       0.911
speed_range_55+ km/h        0.9828      0.042     23.228      0.000       0.900       1.066
seatbelt_none               0.5488      0.044     12.579      0.000       0.463       0.634
airbag_unavail              0.6686      0.068      9.793      0.000       0.535       0.802
===============================================================================================
```

Fig.36: Logistic Regression Summary (Significant only)

Threshold tuning results (threshold = 0.111)

Training set (confusion matrix):

- TN = **5,771** (73.5%), FP = **1,258** (16.0%)
- FN = **114** (1.45%), TP = **708** (9.02%)

Training metrics:

- **Accuracy:** 0.825
- **Recall (death): 0.861**
- **Precision (death):** 0.360
- **F1:** 0.508

	Accuracy	Recall	Precision	F1
0	0.82525	0.86131	0.36012	0.50789

Fig.37: Train Metric Table (Significant only)

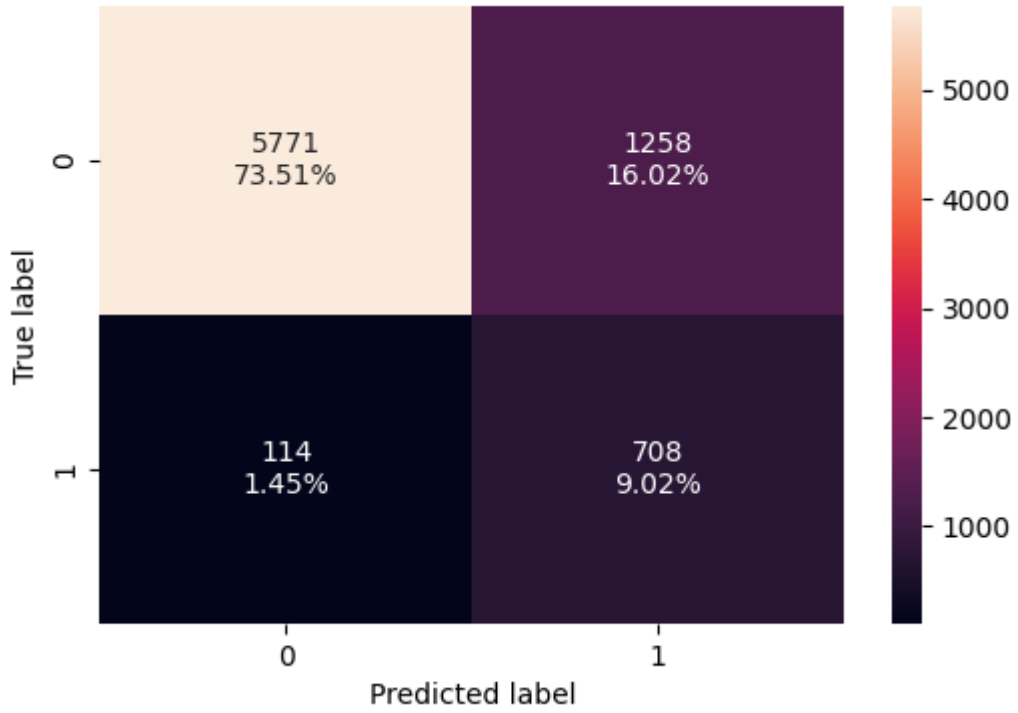


Fig.38: Train Confusion Matrix (Significant only)

Test set (confusion matrix):

- TN = **2,475** (73.5%), FP = **533** (15.8%)
- FN = **52** (1.54%), TP = **306** (9.09%)

Test metrics:

- **Accuracy: 0.826**
- **Recall (death): 0.855**
- **Precision (death): 0.365**
- **F1: 0.511**

	Accuracy	Recall	Precision	F1
0	0.82620	0.85475	0.36472	0.51128

Fig.39: Test Metric Table (Significant only)

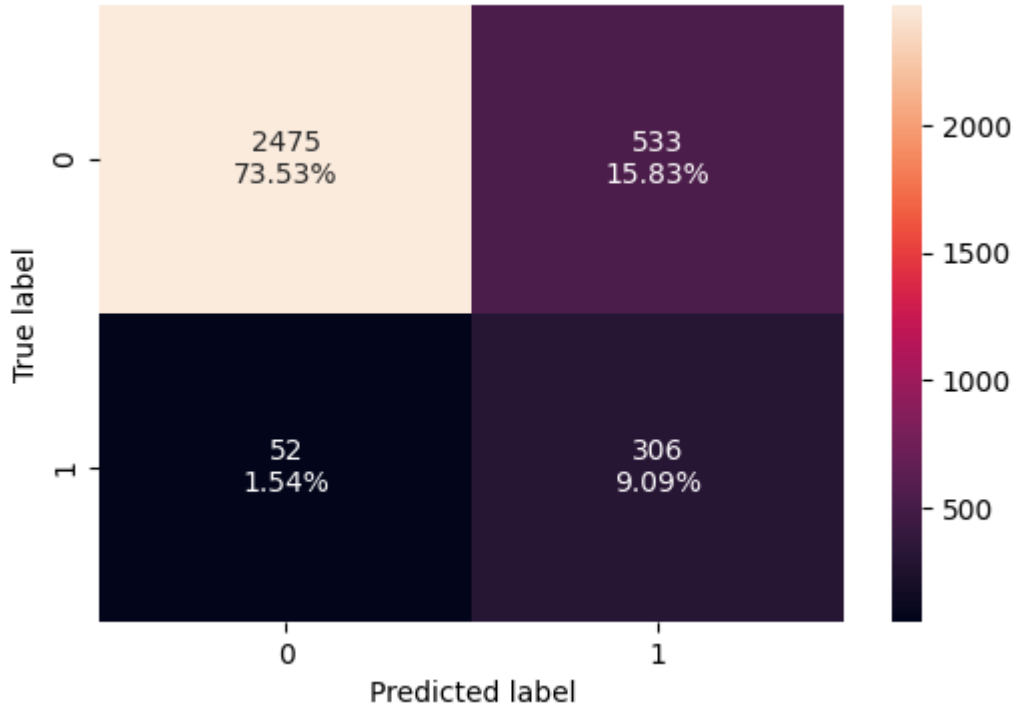


Fig.40: Train Confusion Matrix(Significant only)

How it compares to the baseline LR (0.50 threshold)

- **Recall:** jumped from **0.356** → **0.855**
- **F1:** improved from **0.458** → **0.511**
- **Precision:** dropped **0.640** → **0.365** (expected trade-off)
- **Accuracy:** dropped **0.912** → **0.826** (also expected with a minority-positive policy)

Interpretation for the business

- With a public-safety objective, **missing a fatal case (FN)** is costlier than flagging a non-fatal case (FP).
- The tuned model **captures ~85% of deaths**, making it well-suited as a **risk-screening tool** to trigger interventions (speed enforcement, safety audits, targeted messaging).
- The increased FPs are acceptable in this context; they identify crashes with risk characteristics (high speed, unbelted, old vehicle) even if the outcome was survival.

Decision Tree — Pre-Pruned (Tuned) Model

Objective. Reduce the baseline tree's overfitting and optimize for **recall of fatalities** (minority class) as required by the rubric.

Tuning approach (pre-pruning)

A grid search with 5-fold CV was run on `DecisionTreeClassifier` with the scoring metric set to **recall**. The grid covered:

- `max_depth`: {5, 7, 9, 11, 13}
- `max_leaf_nodes`: {10, 20, 40, 50, 75, 100}
- `min_samples_split`: {2, 5, 7, 10, 20, 30}
- `class_weight`: {'balanced', None}

Best estimator (from CV):
`DecisionTreeClassifier(class_weight='balanced', max_depth=5, max_leaf_nodes=10, random_state=42)`
(*`min_samples_split` fell back to the default because the best combo used the default value.*)

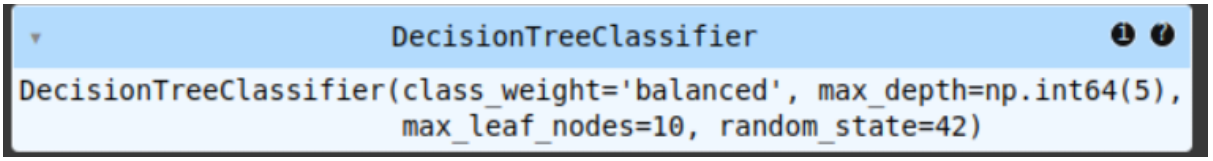


Fig.41: Grid Search output

Generalization performance

Training set

- **Accuracy:** 0.776
- **Recall (death): 0.870**
- **Precision (death):** 0.302
- **F1:** 0.449
- CM: **TN=5,378 (68.5%) / FP=1,651 (21.0%) / FN=107 (1.36%) / TP=715 (9.11%)**

Test set

- **Accuracy: 0.777**
- **Recall (death): 0.855**
- **Precision (death): 0.305**
- **F1: 0.449**
- CM: **TN=2,310 (68.6%) / FP=698 (20.7%) / FN=52 (1.54%) / TP=306 (9.09%)**

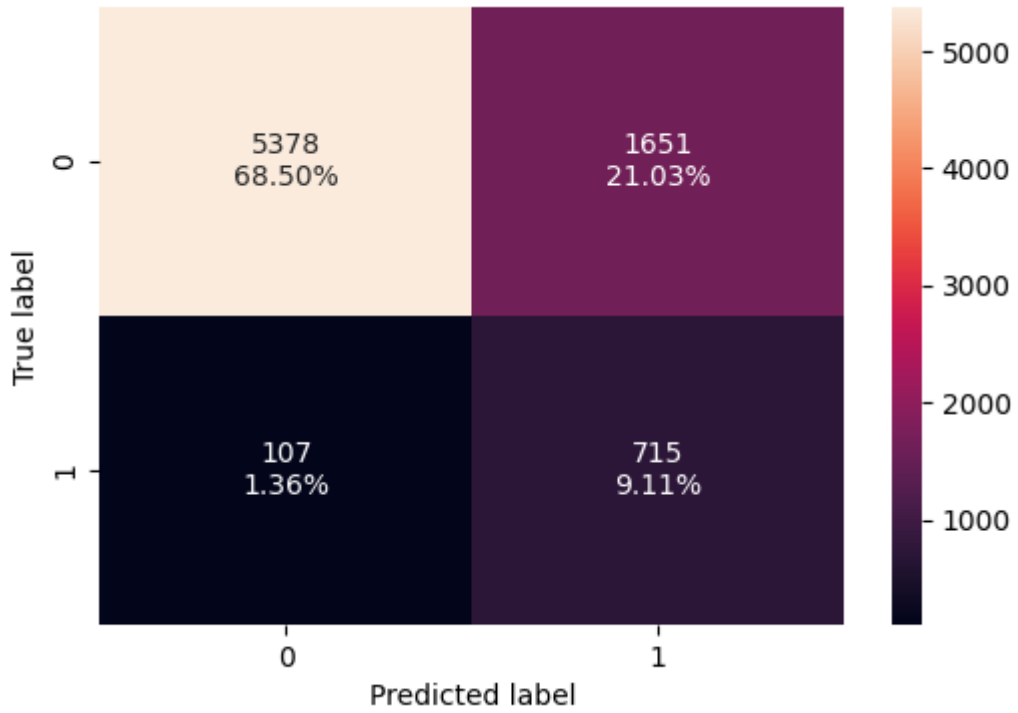


Fig.42: Training confusion matrix

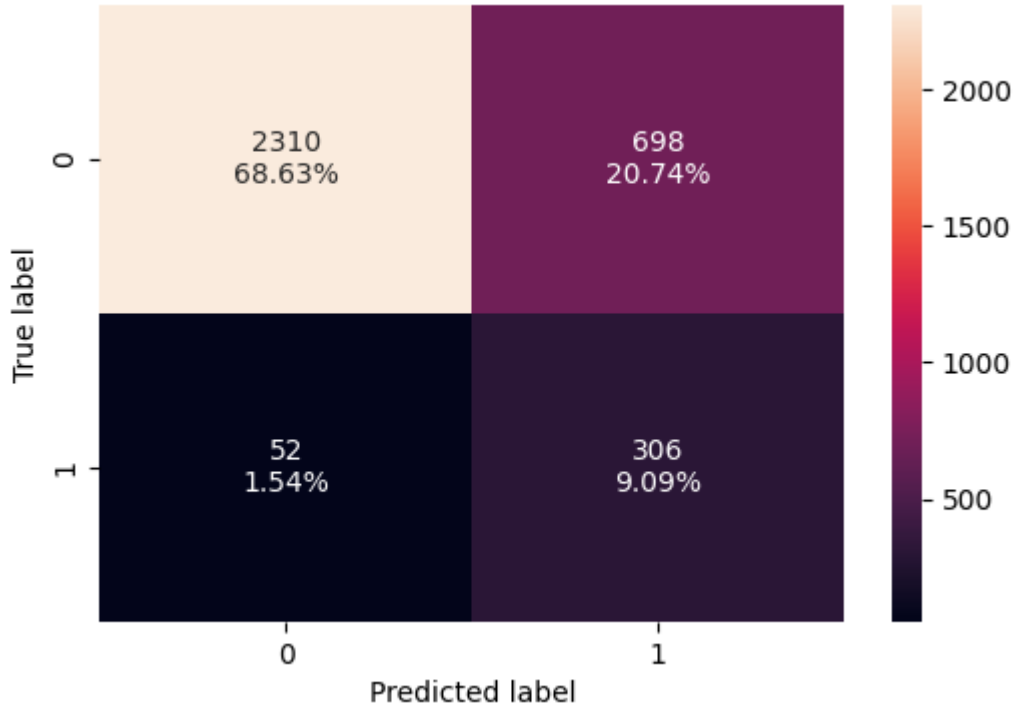


Fig.43: Testing confusion matrix

Interpretation.
Pre-pruning fixed the memorization we saw in the unpruned tree: train and test metrics now **match closely**, so variance is under control. The tuned tree achieves **very high recall (~0.855)**—on par with the tuned logistic model—but at the cost of **lower precision (~0.305)** and **lower accuracy (~0.777)** due to more false positives.

Model explainability

Feature importance (relative):
The tuned tree relies most on:

1. **weight** (largest driver in the tree),
2. **speed_range_10–24 km/h**,
3. **speed_range_55+ km/h**,
4. **speed_range_25–39 km/h**,
5. **age_of_occ**,
6. **seatbelt_none**,
7. **frontal_impact**,

8. **speed_range_40–54 km/h**; the rest contribute minimally.

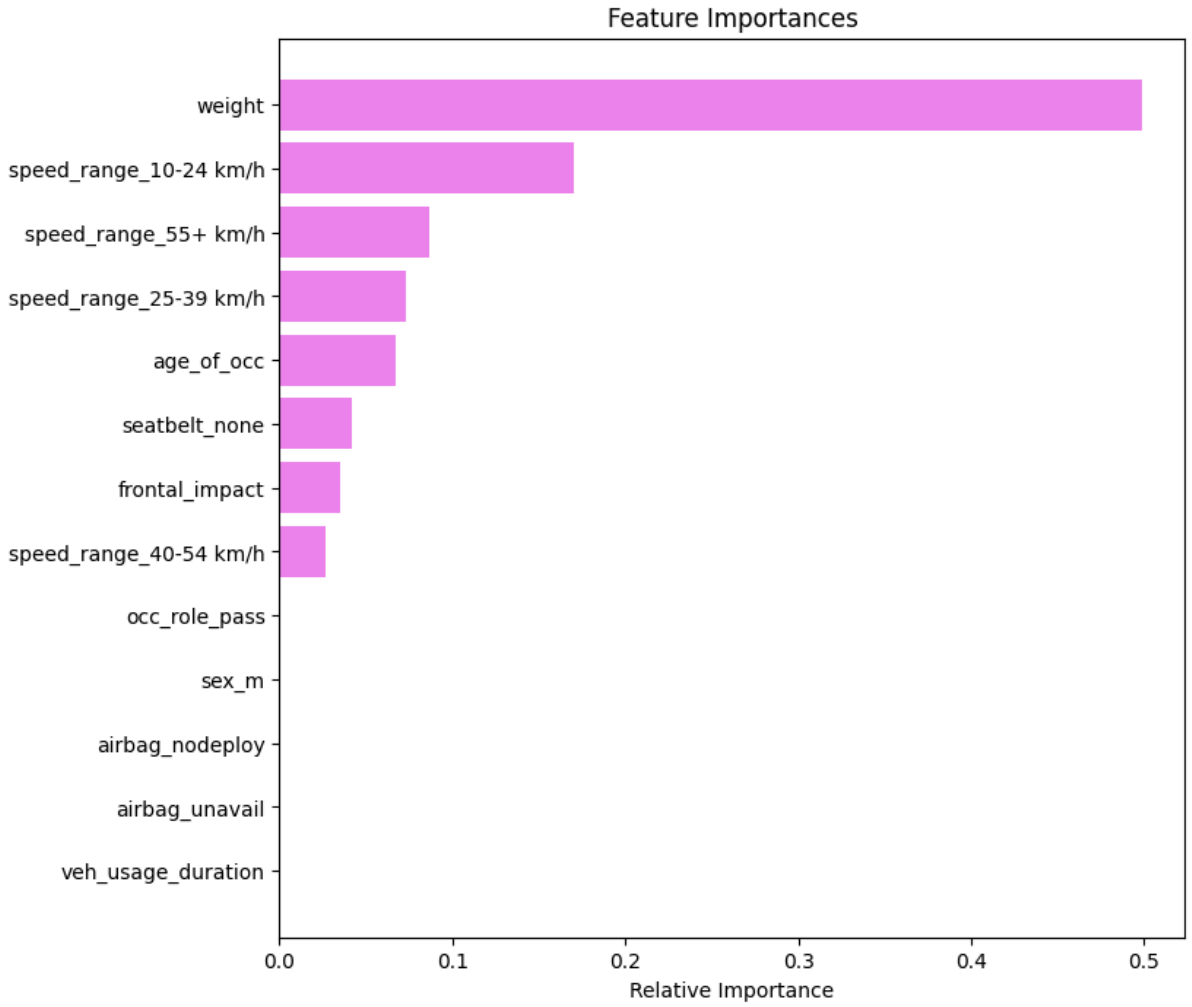


Fig.44: Feature Importance

Tree visualization.
At the root, the split on **weight ≤ ~90.4** partitions the sample, with consequential branches on **speed bands**, **seatbelt** and **age**. This aligns with domain expectations: higher speeds, unbelted occupants, and older people/vehicles push the path toward higher fatality risk leaves.

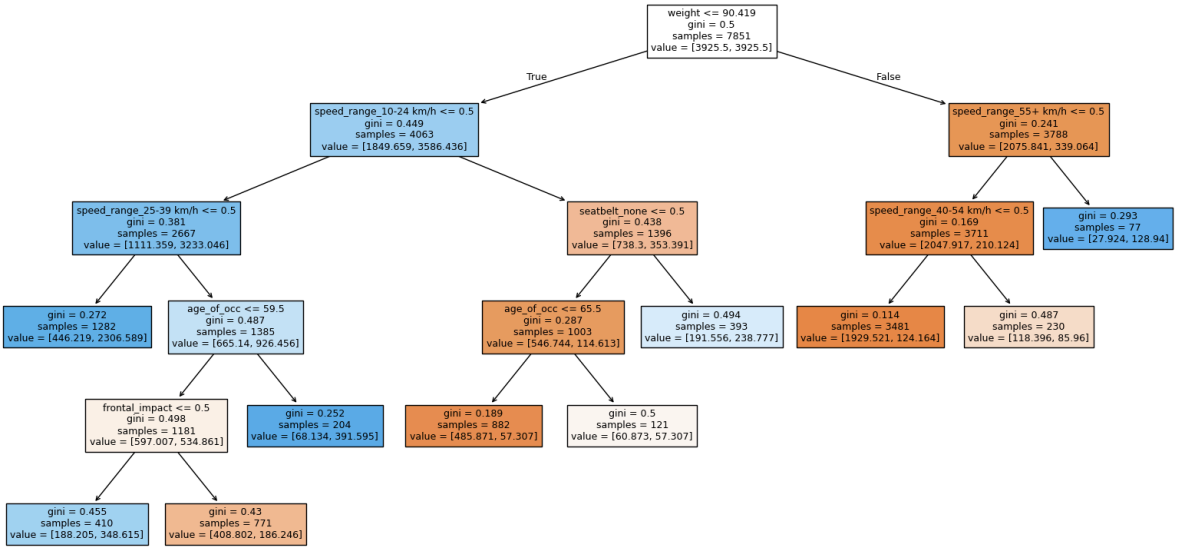


Fig.45: Decision Tree Visualization

Takeaways for the business

- The **pre-pruned decision tree** hits the recall target (captures **~85%** of fatal cases) and is **interpretable**—useful for communicating risk rules (e.g., *older occupants at higher speeds without belts*).
- Compared with the **tuned logistic regression (threshold=0.111)**, this tree has **similar recall** but **lower precision** and **lower accuracy**, meaning more non-fatal crashes are flagged as high risk.
- If the enforcement aim is **maximum detection** with simple rules, this tree is appropriate. If the aim is **fewer false alarms** while keeping high recall, the **tuned logistic** remains the stronger operational model.

Model performance comparison & final model selection

Why these metrics?

The business goal is to pre-empt high-risk crashes. Missing a fatality (false negative) is costlier than a false alarm, so we **optimize recall** of the “deceased = yes” class, and then compare **precision, F1, and accuracy** to control collateral alarms.

Head-to-head on the test set

Test set performance comparison:

	Logistic Regression Base	Logistic Regression (Optimal threshold)	Decision Tree Base	Decision Tree Tuned
Accuracy	0.91503	0.82620	0.88770	0.77718
Recall	0.38268	0.85475	0.42458	0.85475
Precision	0.67822	0.36472	0.46914	0.30478
F1	0.48929	0.51128	0.44575	0.44934

Fig.46: Test comparison table as **Tbl MC-Test**

- **Logistic Regression – base (0.50 threshold):** Acc **0.915**, Recall **0.383**, Prec **0.678**, F1 **0.489**
  High accuracy from class imbalance, but recall is too low for the safety objective.
- **Logistic Regression – tuned threshold (θ≈0.111):** Acc **0.826**, **Recall 0.855**, Prec **0.365**, **F1 0.511**
  Huge recall lift with a modest fall in accuracy; best F1 among high-recall models.
- **Decision Tree – base (unpruned):** Acc **0.888**, Recall **0.425**, Prec **0.469**, F1 **0.446**
  Recall still low; underperforms tuned logit on every metric that matters here.
- **Decision Tree – pre-pruned (CV for recall):** Acc **0.777**, **Recall 0.855**, Prec **0.305**, F1 **0.449**
  Matches the tuned logit on recall, but with **lower precision** and **lower accuracy**.

Generalization check on the training set

Training performance comparison:

	Logistic Regression Base	Logistic Regression (Optimal threshold)	Decision Tree Base	Decision Tree Tuned
Accuracy	0.91160	0.82525	1.00000	0.77608
Recall	0.35645	0.86131	1.00000	0.86983
Precision	0.63974	0.36012	1.00000	0.30220
F1	0.45781	0.50789	1.00000	0.44856

Fig.47: Training comparison table as **Tbl MC-Train**.

- The **base decision tree** shows perfect 1.00/1.00/1.00/1.00—classic **overfitting**.
- **Pre-pruning** fixes this: train ≈ test, confirming variance control.
- **Tuned logistic** also shows **stable train↔test metrics**, a good bias-variance trade-off.

Final model choice

Select: Logistic Regression with significant features + threshold tuned via ROC (θ≈0.111).
Why: It delivers the **same best-in-class recall (≈0.855)** as the tuned tree **but with better precision (0.365 vs 0.305), higher F1 (0.511 vs 0.449), and higher accuracy (0.826 vs 0.777)**. That means **fewer false alarms** for the same level of life-saving sensitivity—more practical for enforcement and public adoption.

Where to place figures: Use the tuned-logit **ROC curve** and **confusion matrices** you generated earlier as **Fig LOG-ROC** and **Fig LOG-CM (train/test)** immediately after **Tbl MC-Test** to visually anchor the selection.

What this means operationally

- **Operating point:** Use **θ≈0.111** for deployment; it maximizes TPR−FPR on ROC.
- **Expected outcomes on new data (based on test):**
  – **~85.5%** of fatal cases will be flagged (high recall).
  – For each predicted fatality, **~36%** are true fatal cases (precision).
  – Overall accuracy around **82–83%** despite class imbalance.

- **Risk policy:** If stakeholders want even **fewer false positives**, raise θ slightly; if they want **near-max capture** of fatalities at the expense of more alerts, lower θ. The ROC lets regulators pick the point matching their cost tolerance.

Model transparency & drivers (tie-back)

From the earlier coefficient/importance analyses:

- **Higher speeds**, **no seatbelt**, **older occupants**, and **lower vehicle weight** increase fatality risk; **longer vehicle usage** also raises risk (proxy for wear).
- These patterns are consistent across both models, increasing confidence in policy directions (e.g., belt enforcement, speed management, minimum safety mass, vehicle age/inspection norms).

Actionable Insights & Recommendations

Actionable Insights

1. **Speed is the strongest predictor of fatalities**
   Fatality rates rise sharply with crash speed, exceeding 50% at 55+ km/h. While most crashes occur at lower speeds (10–39 km/h), the deadliest outcomes cluster at high speeds.
2. **Seatbelt use remains inconsistent and critical**
   Occupants without seatbelts face over **3× higher fatality risk** (20.2% vs 6.4%). Despite regulations, nearly 30% of crash occupants were unbelted, showing a persistent compliance gap.
3. **Vehicle age significantly affects survival**
   Older vehicles (longer *veh_usage_duration*) show disproportionately higher fatality rates due to weaker structures and lack of airbags. Survivors are more concentrated in newer cars (<5 years old).

4. **Non-frontal impacts are more deadly than frontal**
   Even though frontal crashes are more frequent (64%), non-frontal crashes show **almost double the fatality rate** (15% vs 8.1%). This highlights the limitations of safety systems optimized for frontal impacts.

Recommendations

1. **Enforce and incentivize speed management policies**
   Deploy stricter speed checks in urban areas, expand use of speed cameras, and consider variable speed limits in high-risk zones.
2. **Targeted seatbelt compliance campaigns**
   Pair stricter fines with public awareness campaigns, focusing on high-risk demographics (young drivers, urban commuters). Linking insurance benefits to consistent seatbelt use could reinforce compliance.
3. **Vehicle renewal and safety upgrade programs**
   Encourage phasing out older vehicles (>15 years) through scrappage incentives, mandatory safety inspections, or retrofitted safety features (side airbags, reinforced structures).